

Introduction



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Objectives

- **Understand** the main stages of IC design flows
- **Study** core data structures and algorithms for Electronic Design Automation (EDA)
- **Formulate** mathematical models for complex optimization problems
- **Evaluate** trade-offs between QoR and computational complexity
- **Transfer** acquired optimization techniques to other engineering domains

Organization

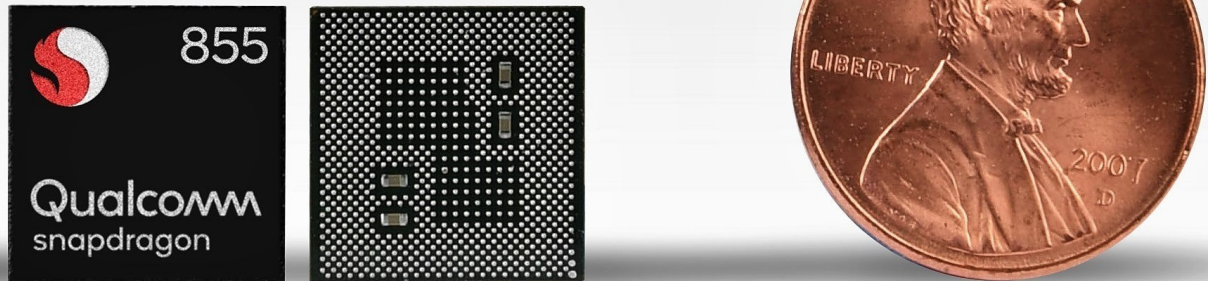
- Lectures: Mon, Thu (12:00-14:00)
- Evaluation:
 - Exercises:
 - physical synthesis (15%)
 - logic synthesis (15%)
 - Course project: EDA tool (35%)
 - One or two students per project
 - Presentation at the end of the course
 - Final exam (35%)
 - A reduced set of notes allowed (few pages on paper)

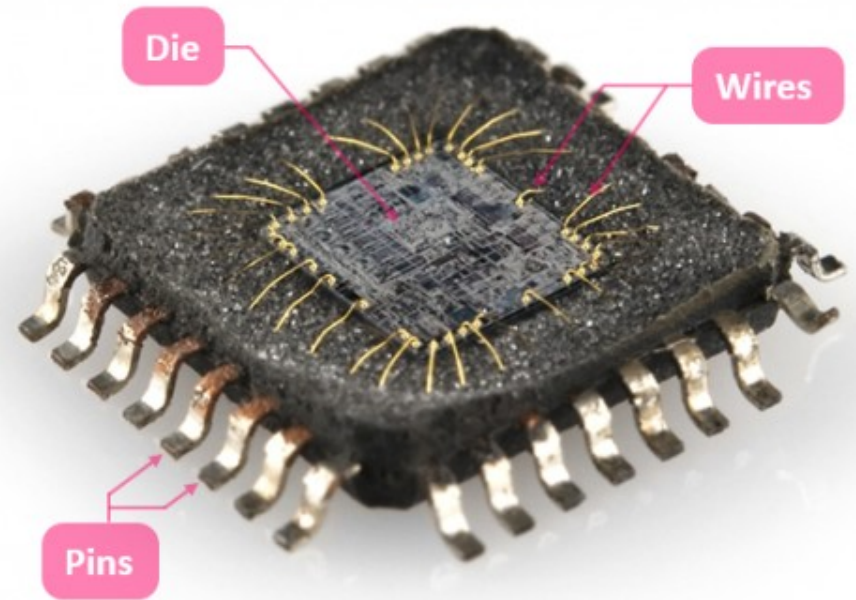
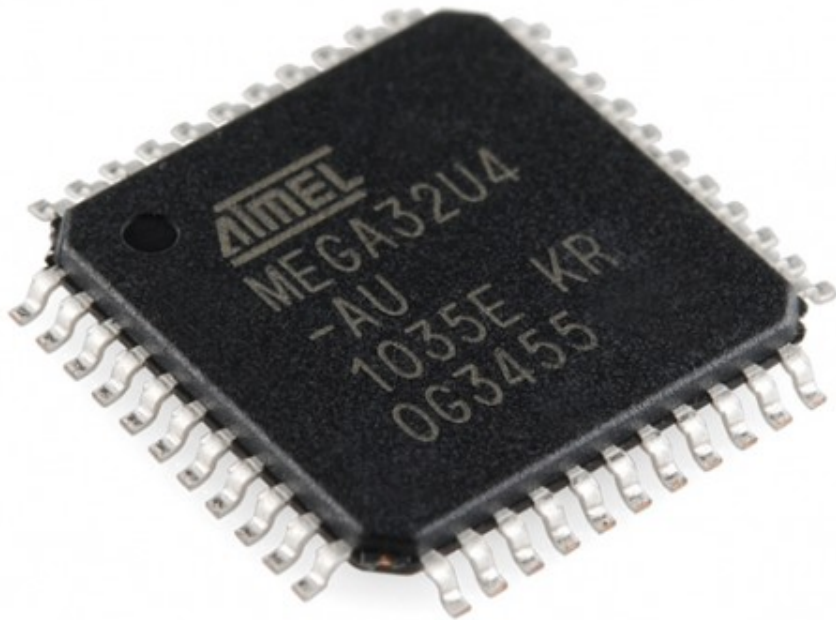
Projects

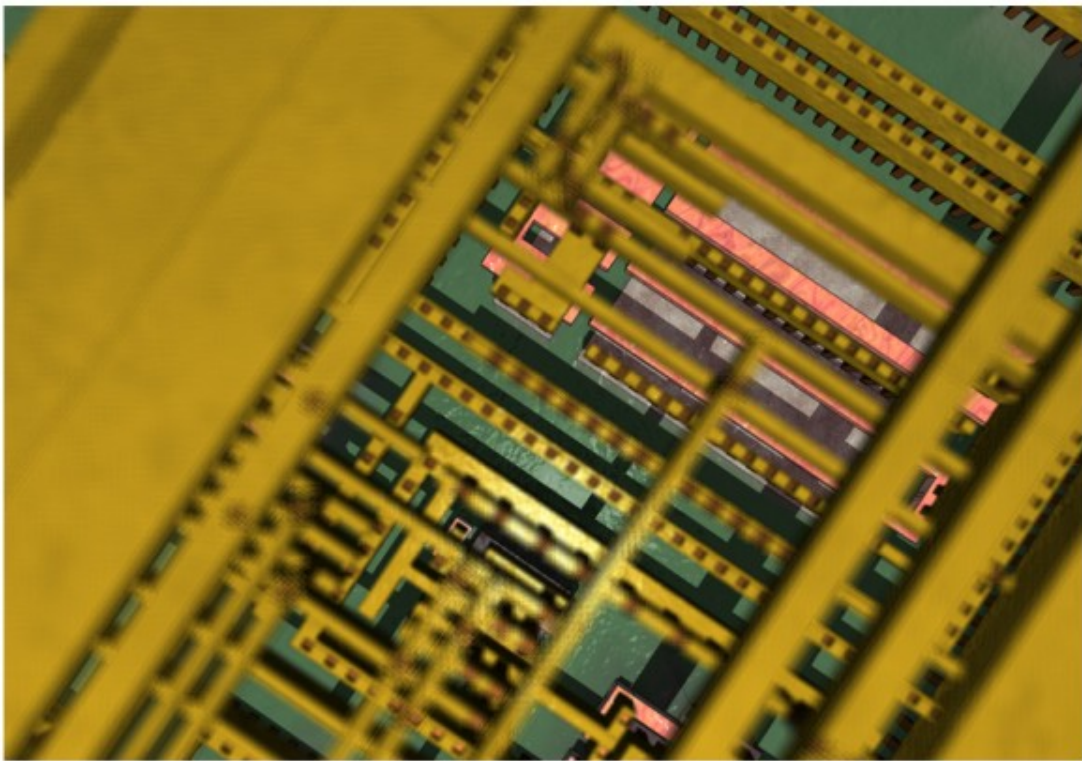
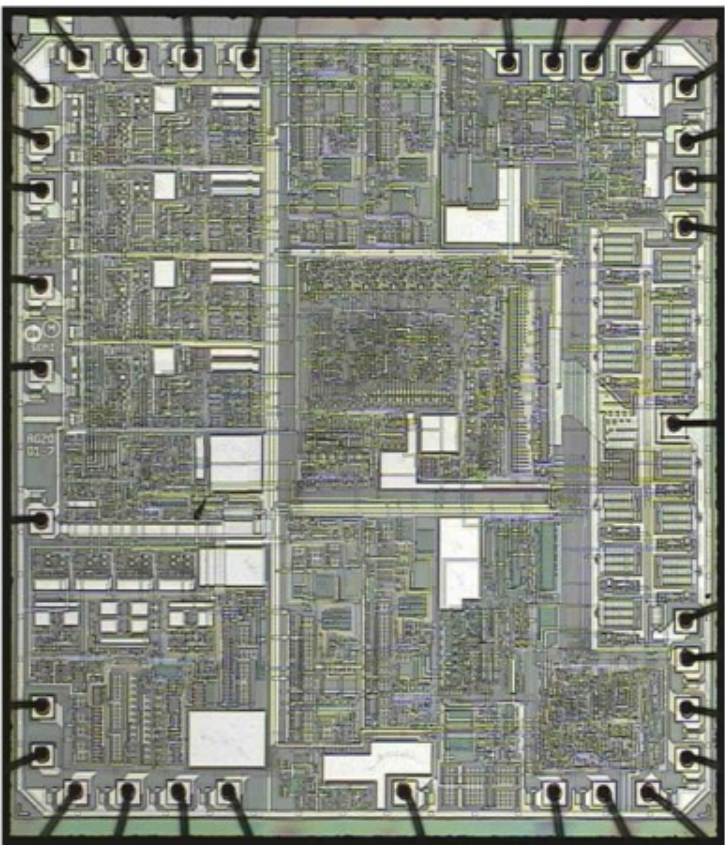
- Graph drawing
- Floorplanning
- Cell placement
 - Global placement
 - Detailed placement
- Routing:
 - Channel
 - Global
 - Detailed
- Boolean minimization (Quine-McCluskey)
- Algebraic factorization
- Any algorithmic challenge proposed by the students (to be discussed)

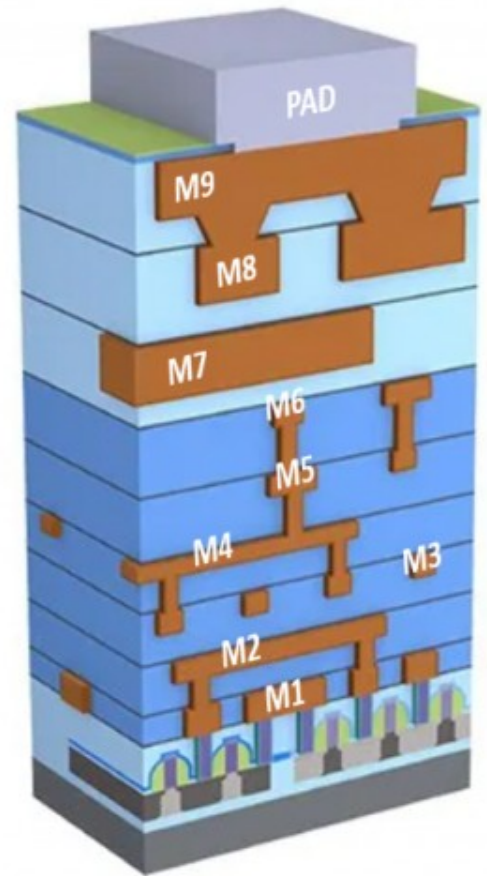
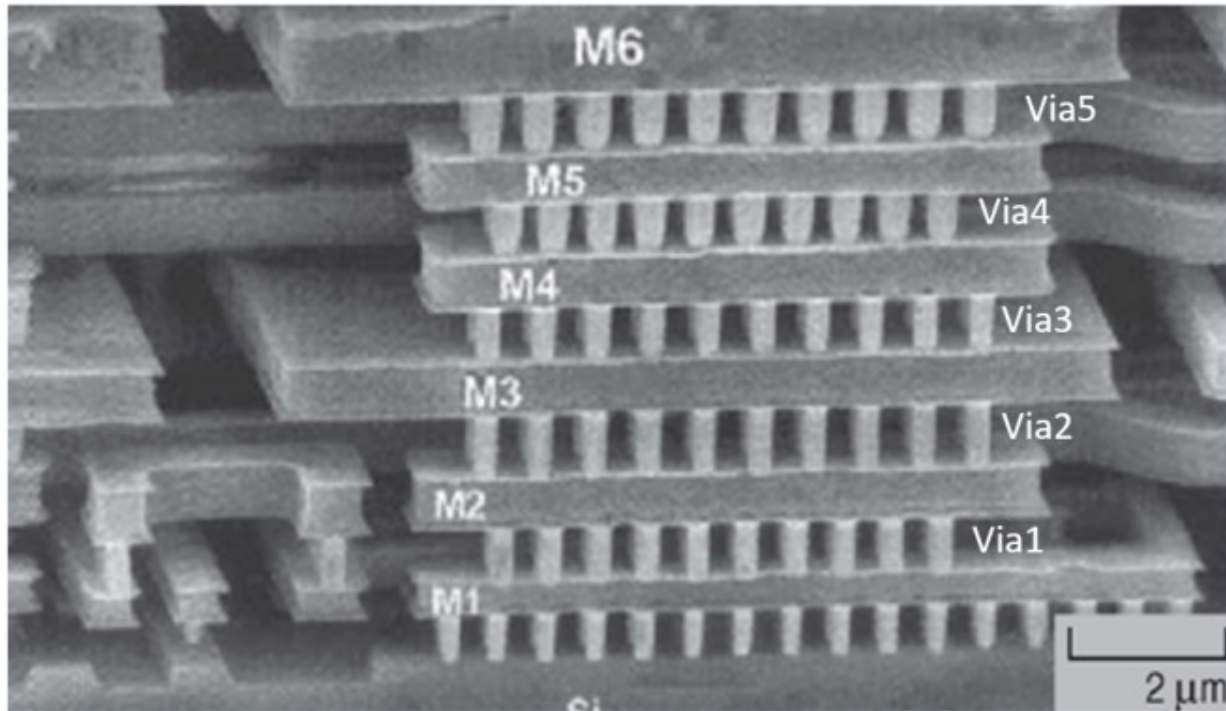
PUTTING PARIS IN A BOTTLE

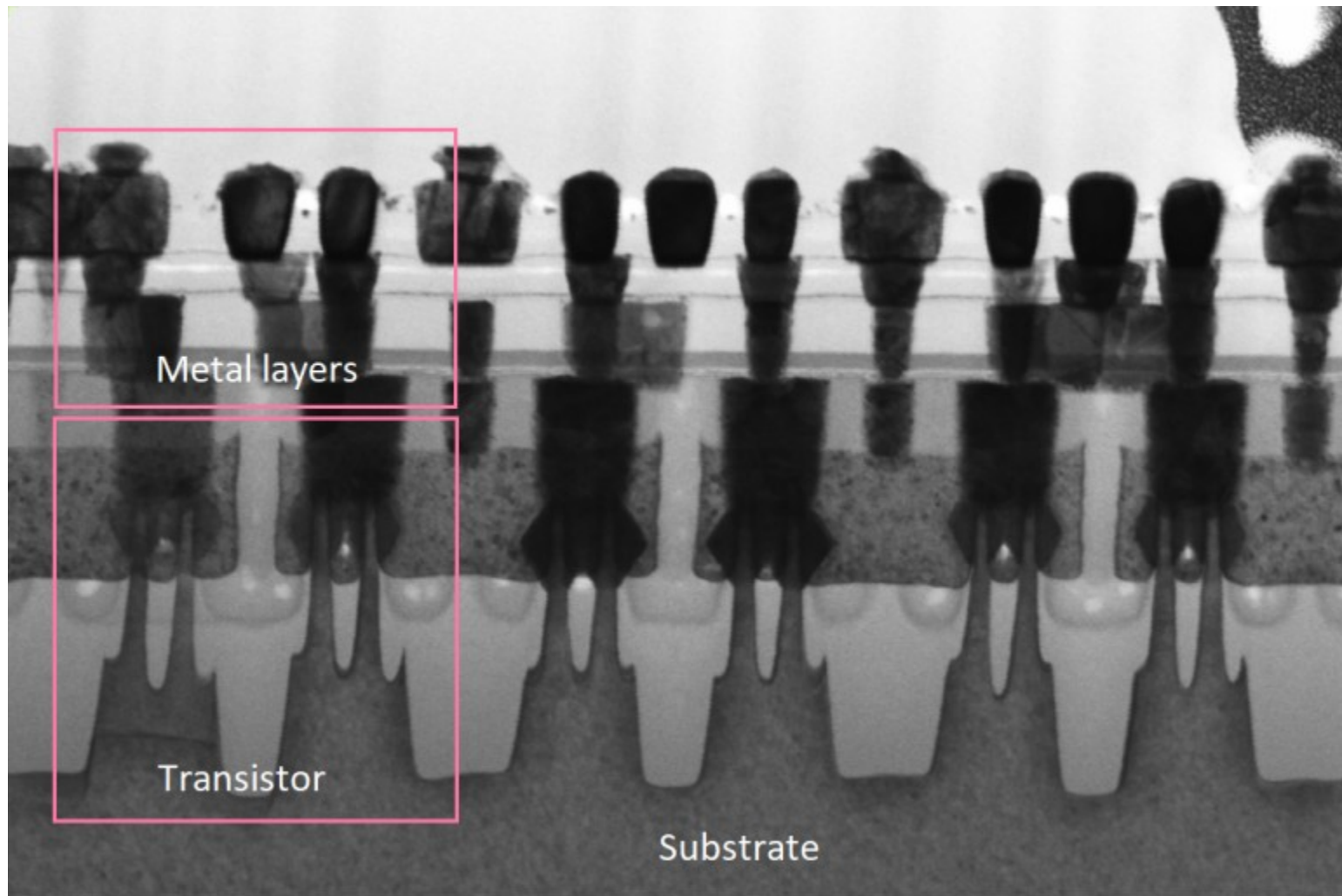
Electronics / By admin / May 20, 2020

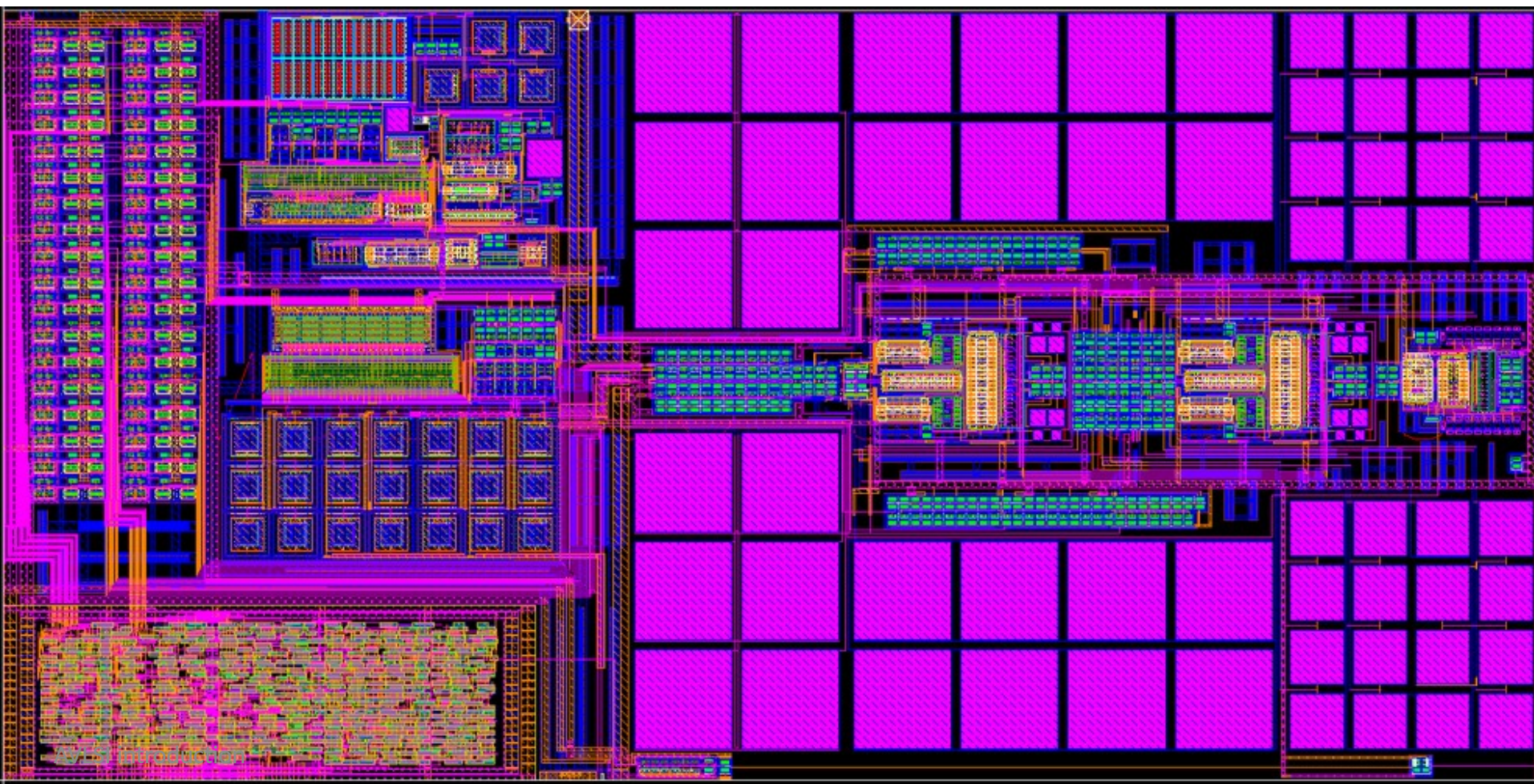
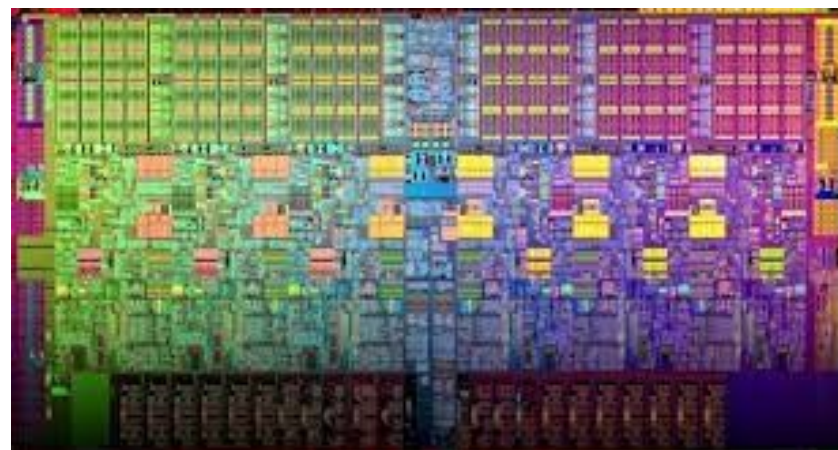
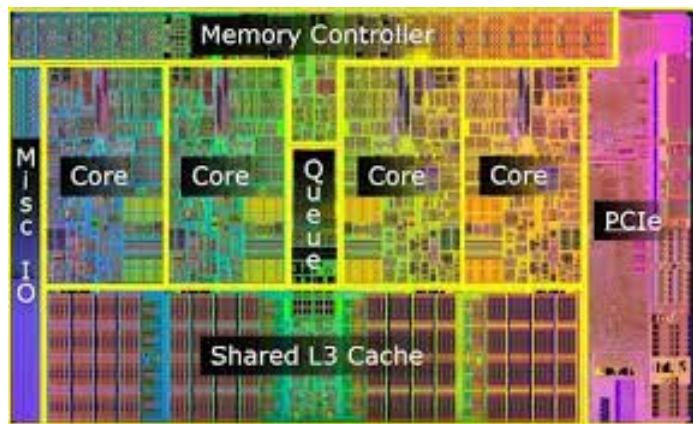








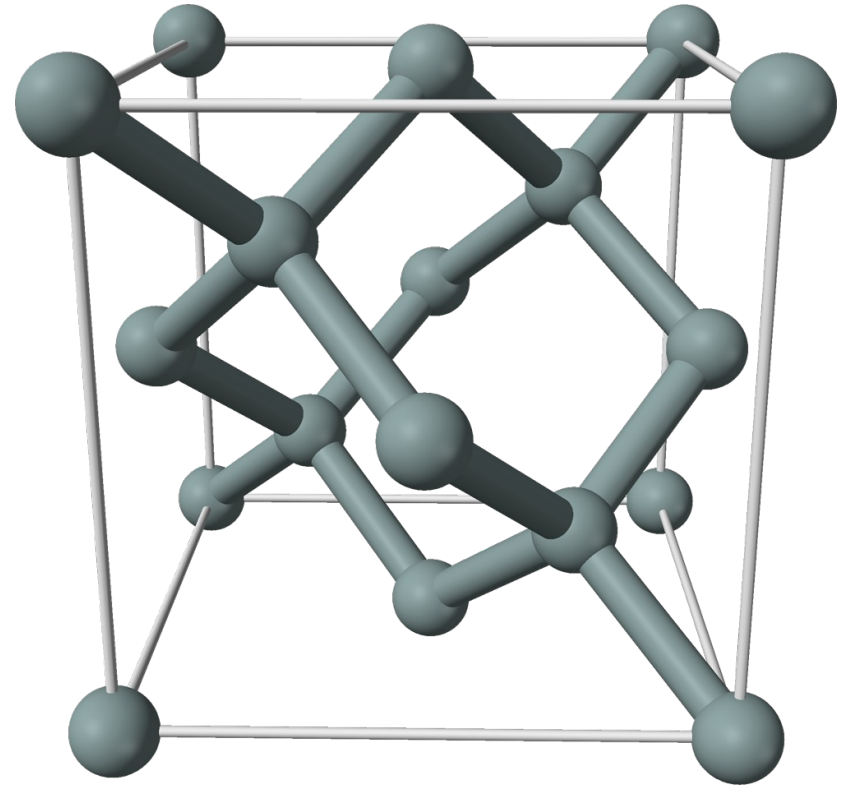
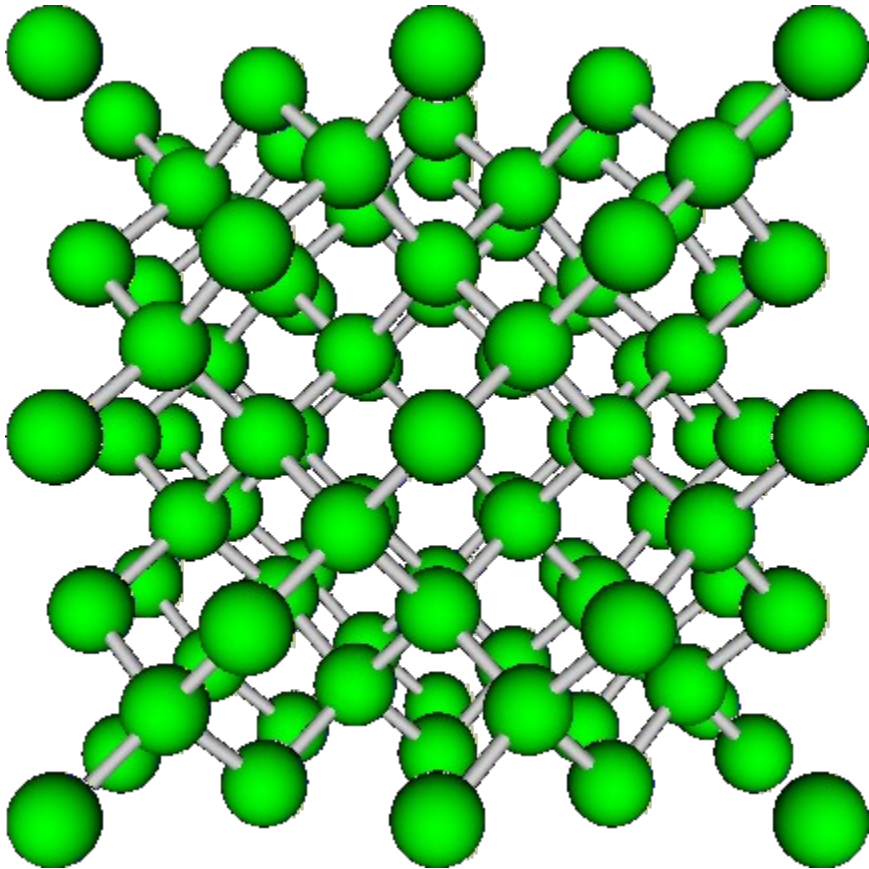




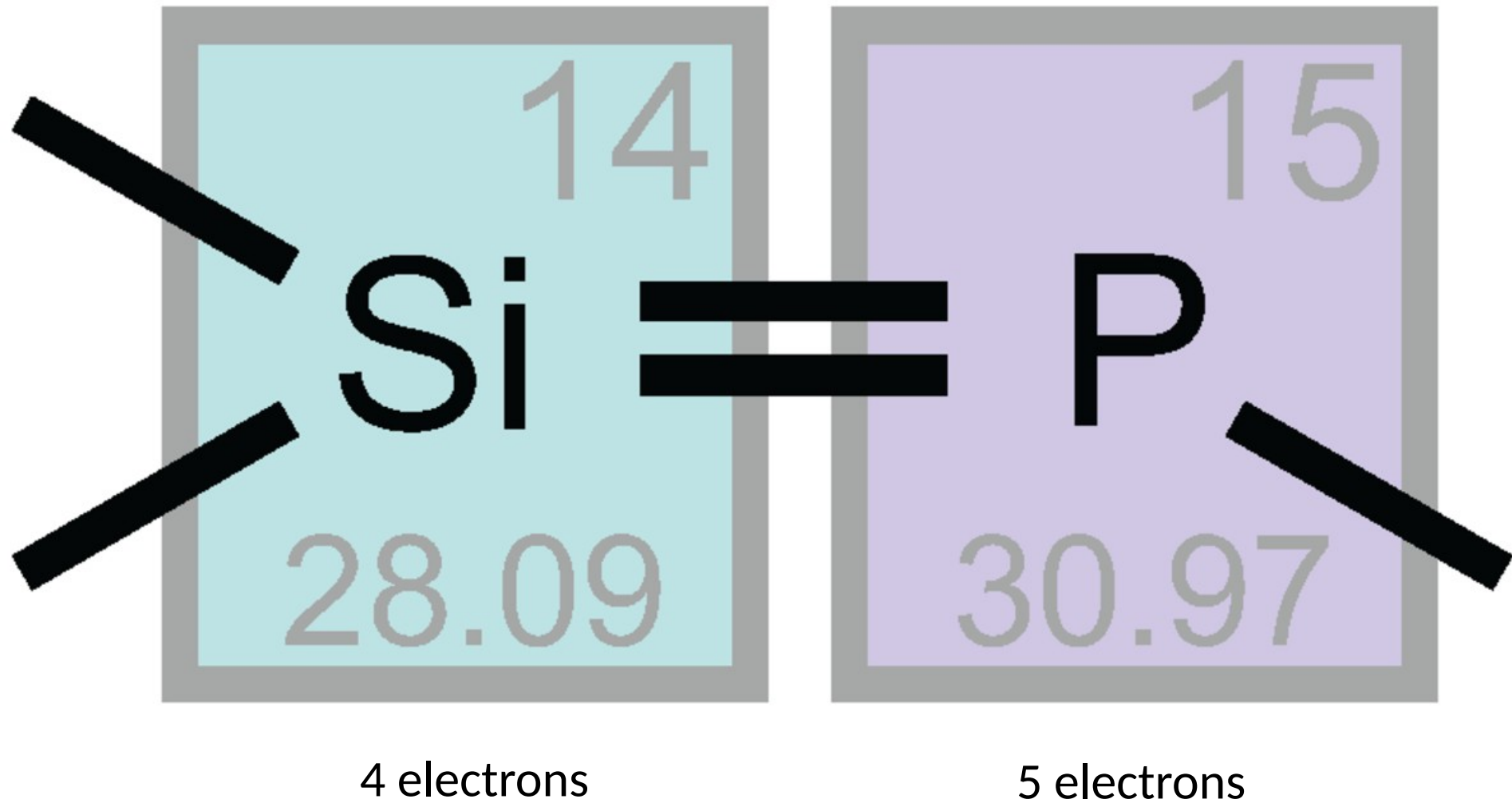
Outline

- How transistors are built and work
- How logic circuits are designed
- How masks are generated for photolithographic processes

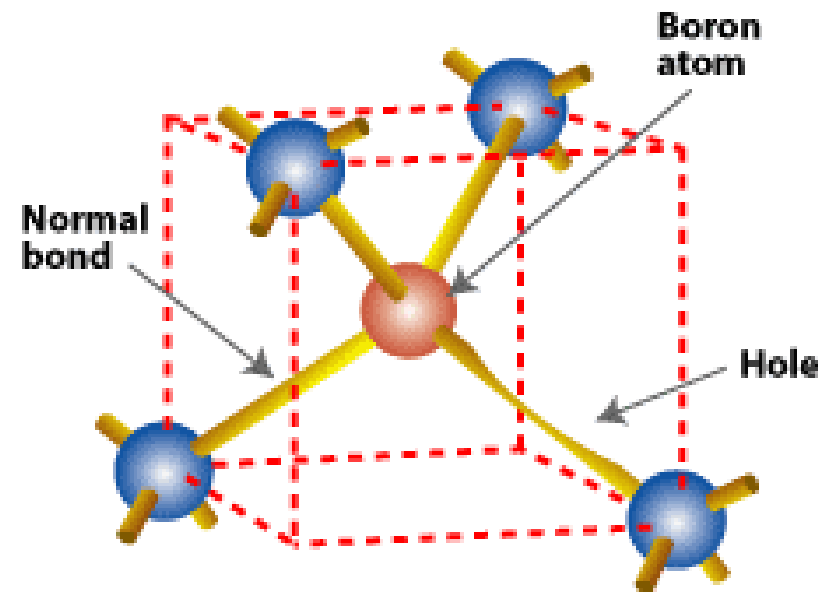
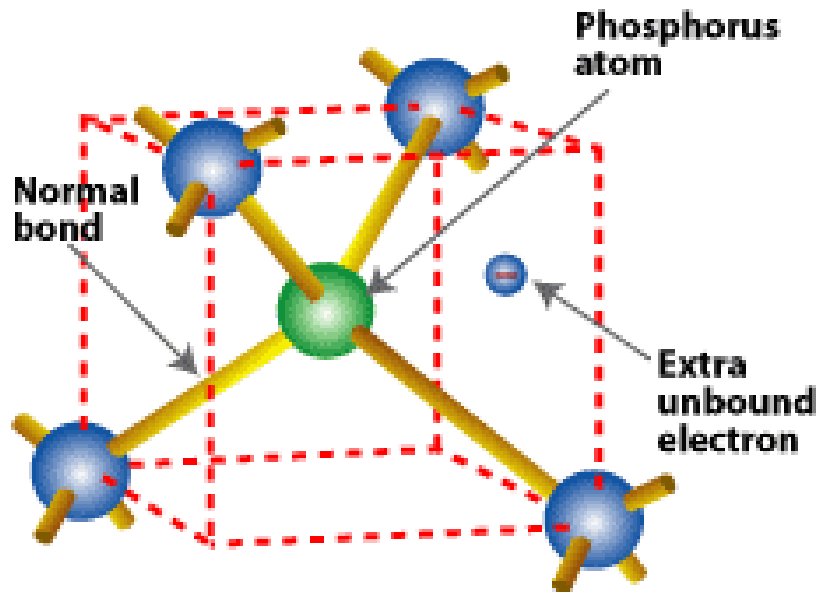
Crystalline silicon



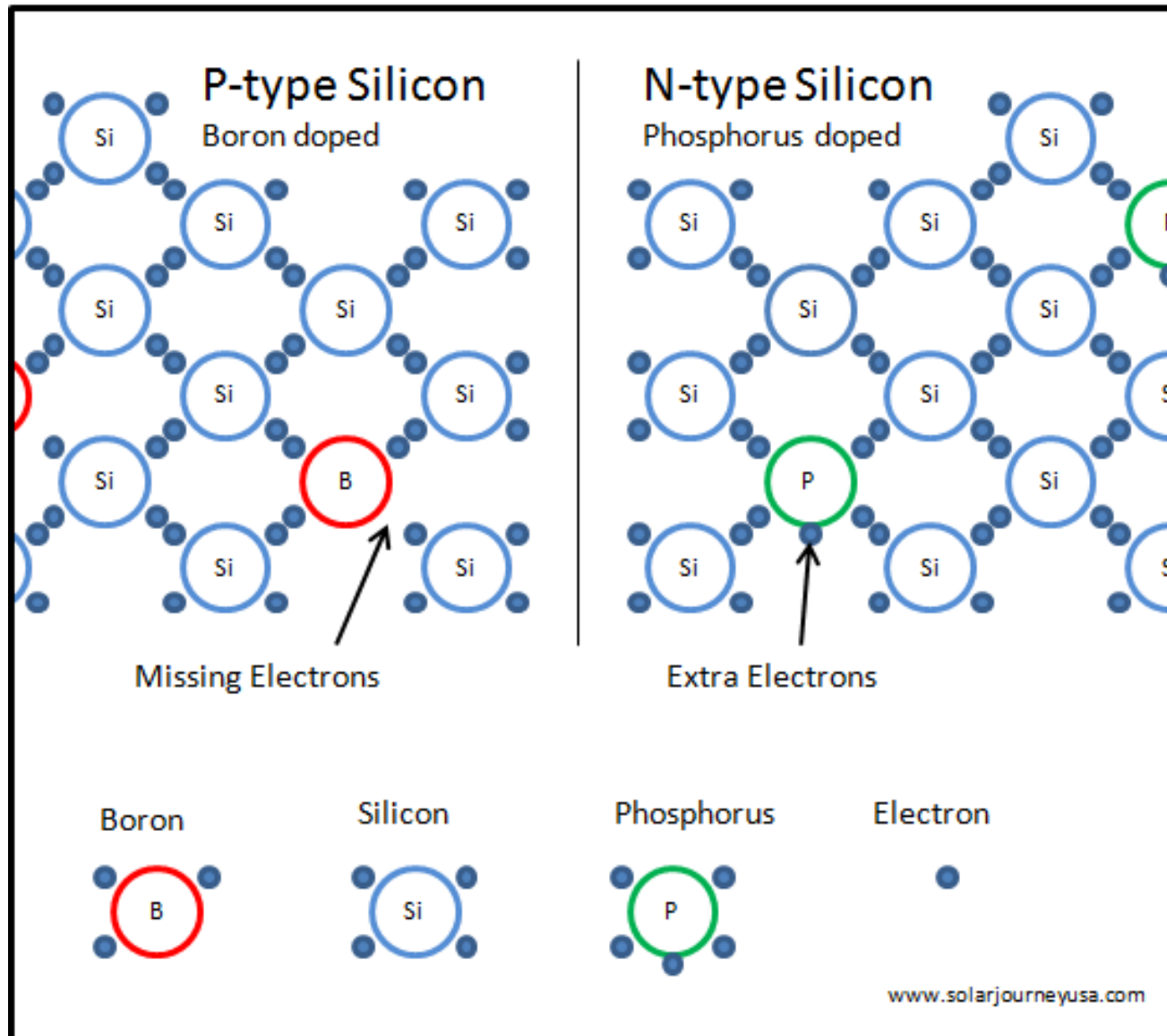
Silicon and Phosphorus: similar size



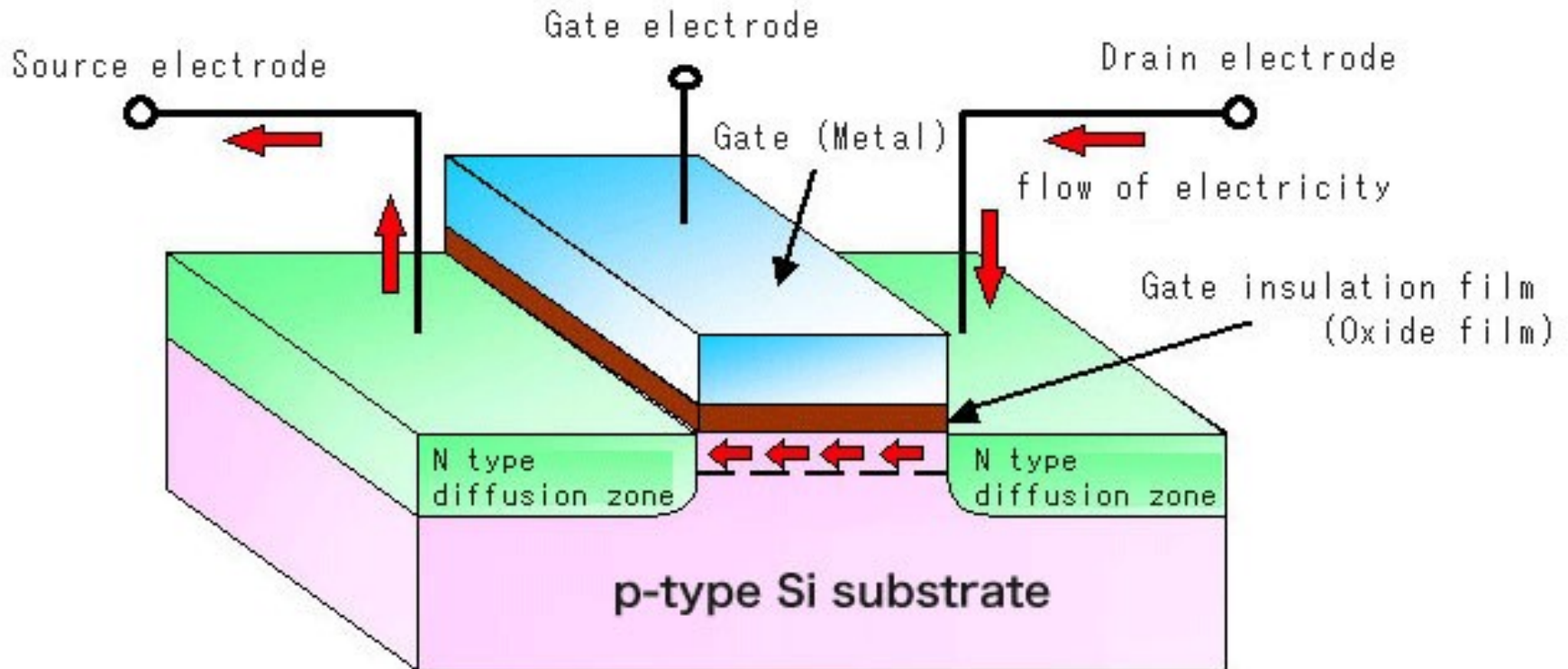
Doped silicon: Phosphorus and Boron

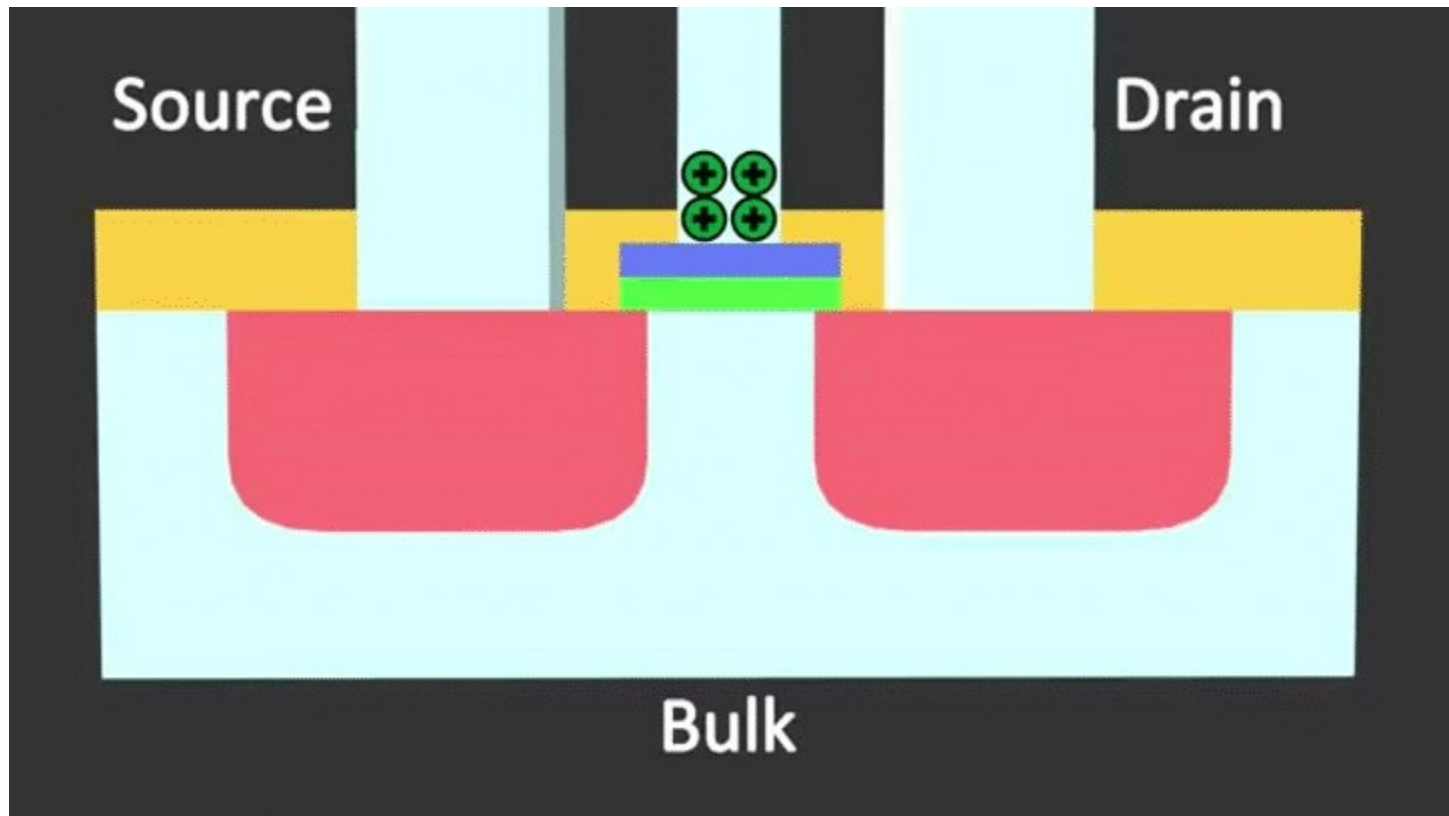


Doped silicon

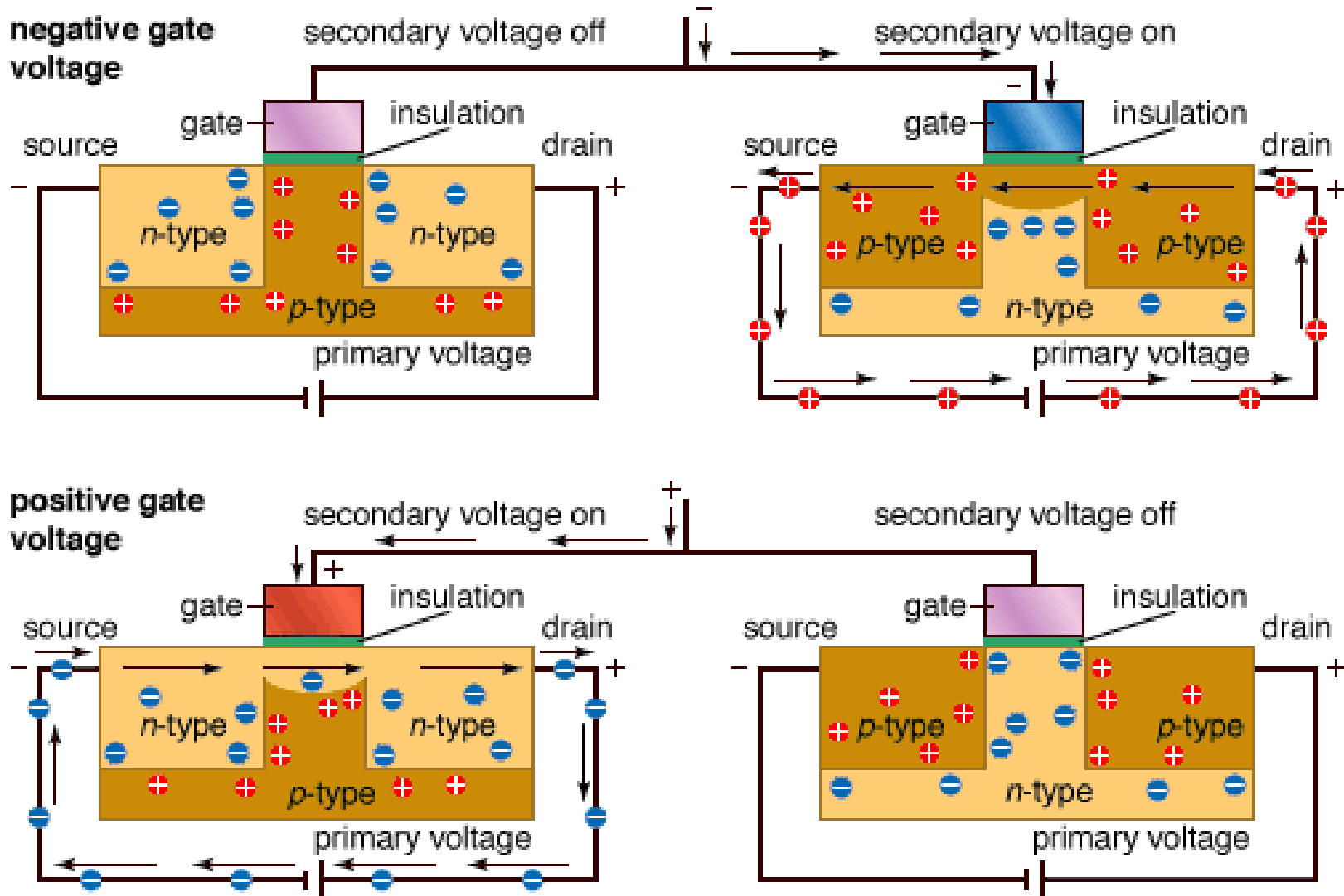


n-type transistor



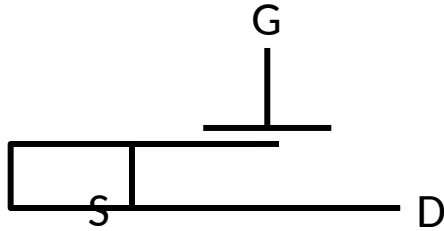


Note: be patient and wait for the animation

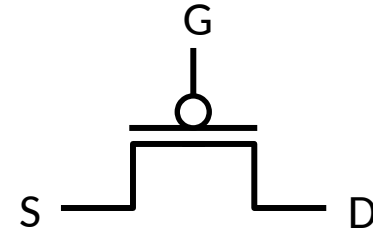


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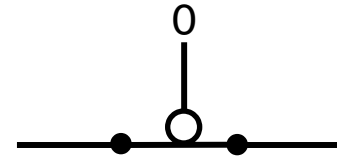
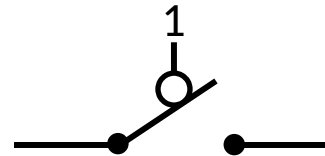
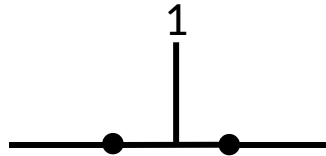
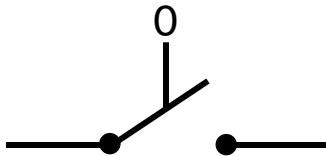
Transistor as a switch

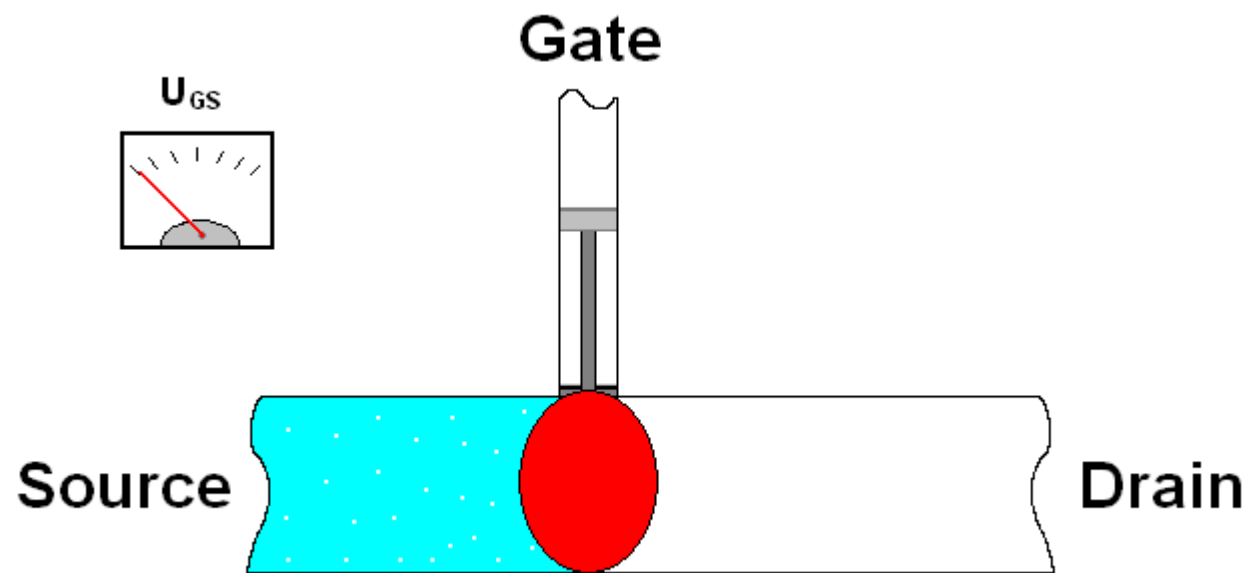


n-type

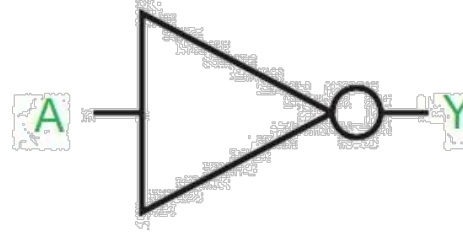
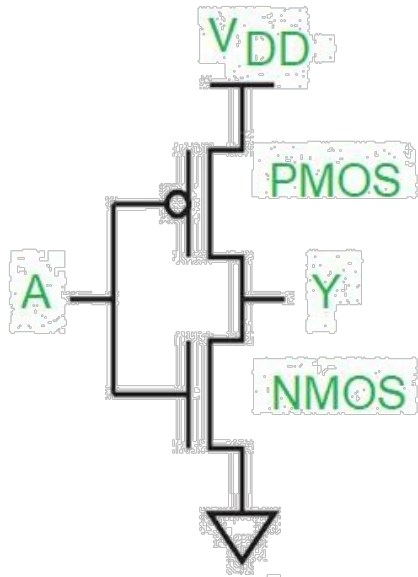


p-type

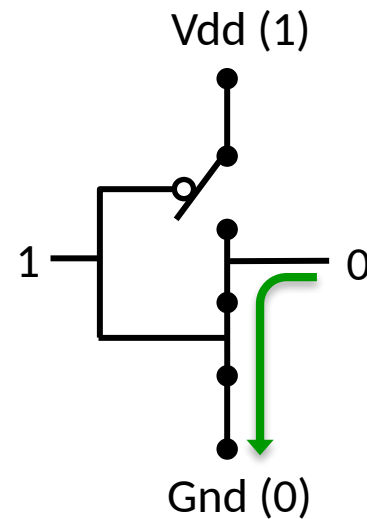
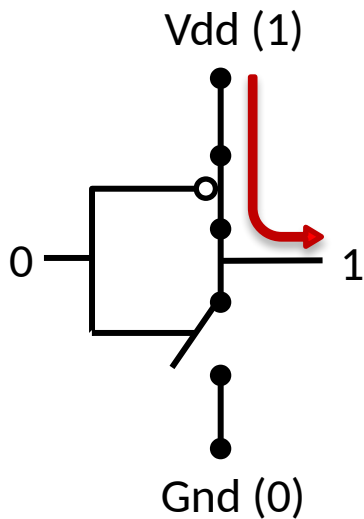




The inverter (NOT)



A	Y
0	1
1	0

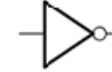


YES



INPUT		OUTPUT
A		
0		0
1		1

NOT



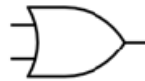
INPUT		OUTPUT
A		
0		1
1		0

AND



INPUT		OUTPUT
A	B	
0	0	0
1	0	0
0	1	0
1	1	1

OR



INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	1

XOR



INPUT		OUTPUT
A	B	
0	0	0
1	0	1
0	1	1
1	1	0

NAND



INPUT		OUTPUT
A	B	
0	0	1
1	0	1
0	1	1
1	1	0

NOR

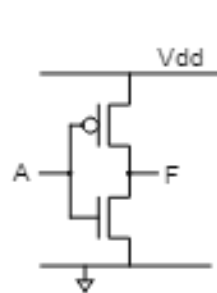


INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	0

XNOR

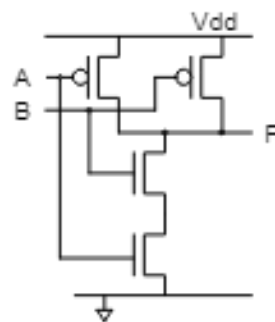


INPUT		OUTPUT
A	B	
0	0	1
1	0	0
0	1	0
1	1	1



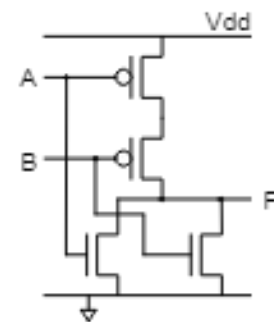
A	F
L	H
H	L

CMOS INVERTER



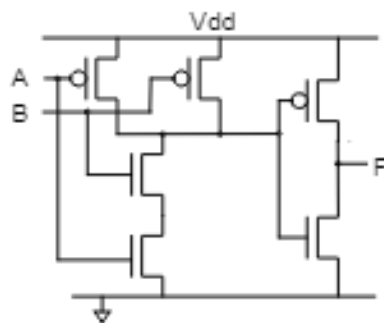
A	B	F
L	L	H
L	H	H
H	L	H
H	H	L

CMOS NAND



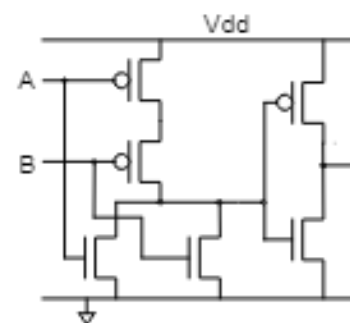
A	B	F
L	L	H
L	H	L
H	L	L
H	H	L

CMOS NOR



A	B	F
L	L	L
L	H	L
H	L	L
H	H	H

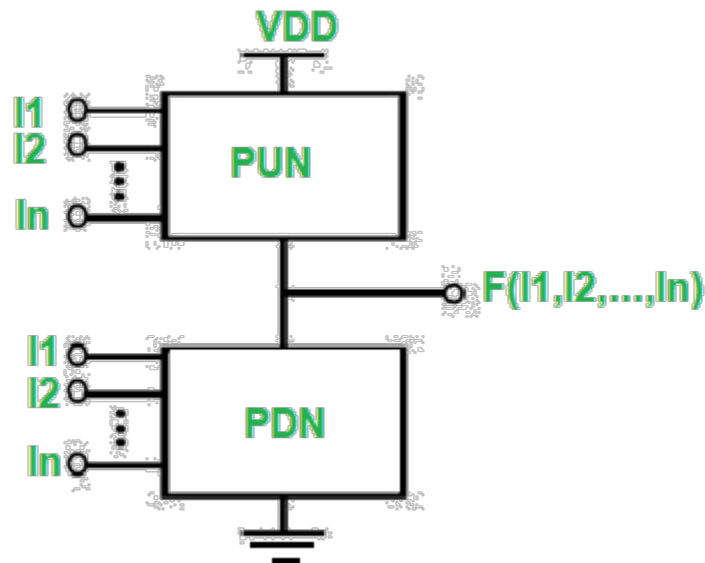
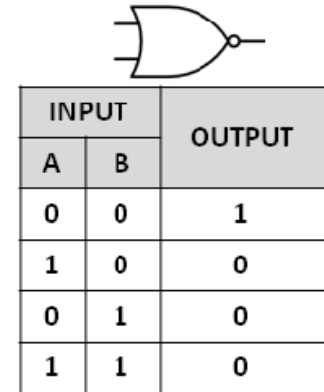
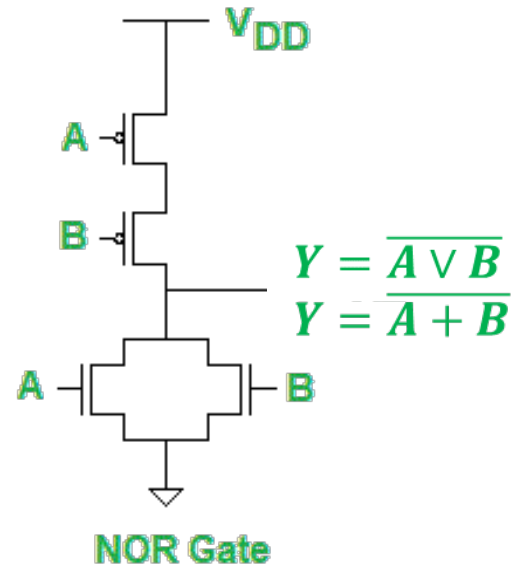
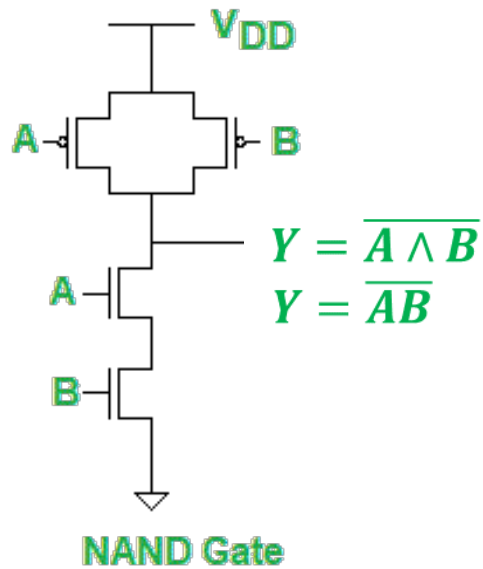
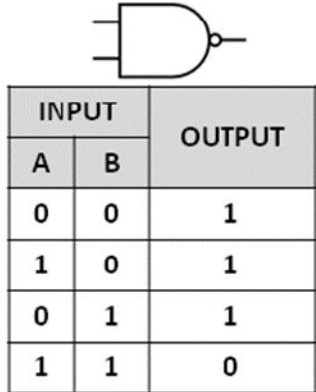
CMOS AND

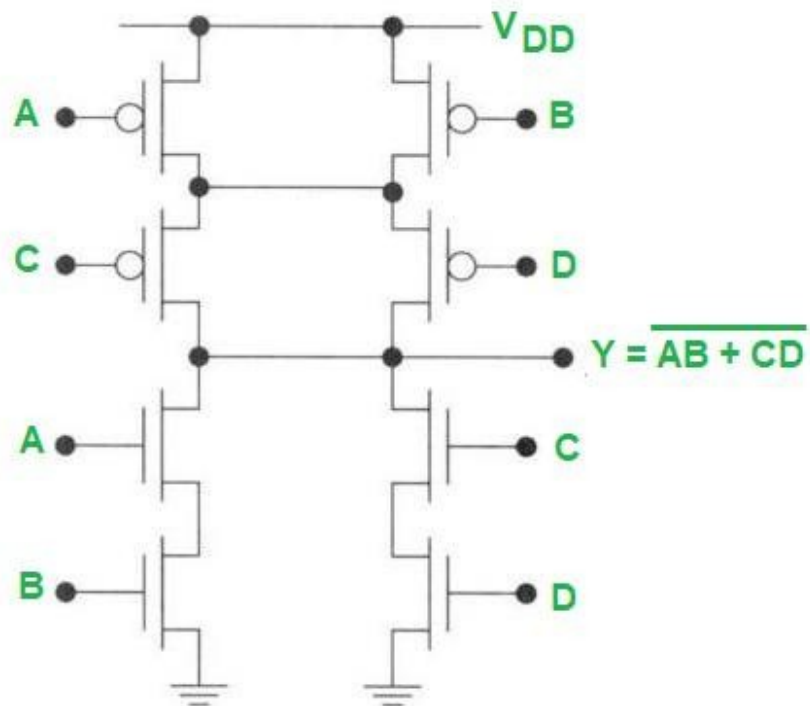
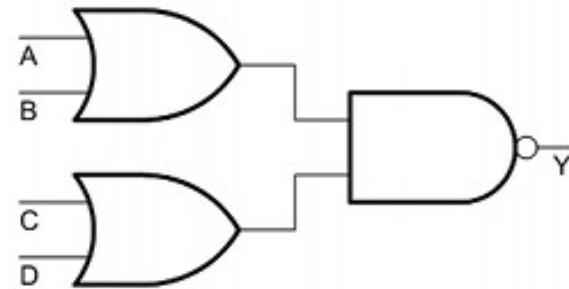
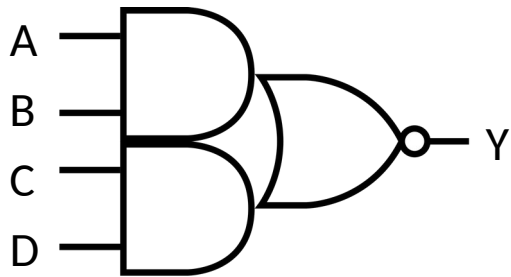


A	B	F
L	L	L
L	H	H
H	L	H
H	H	H

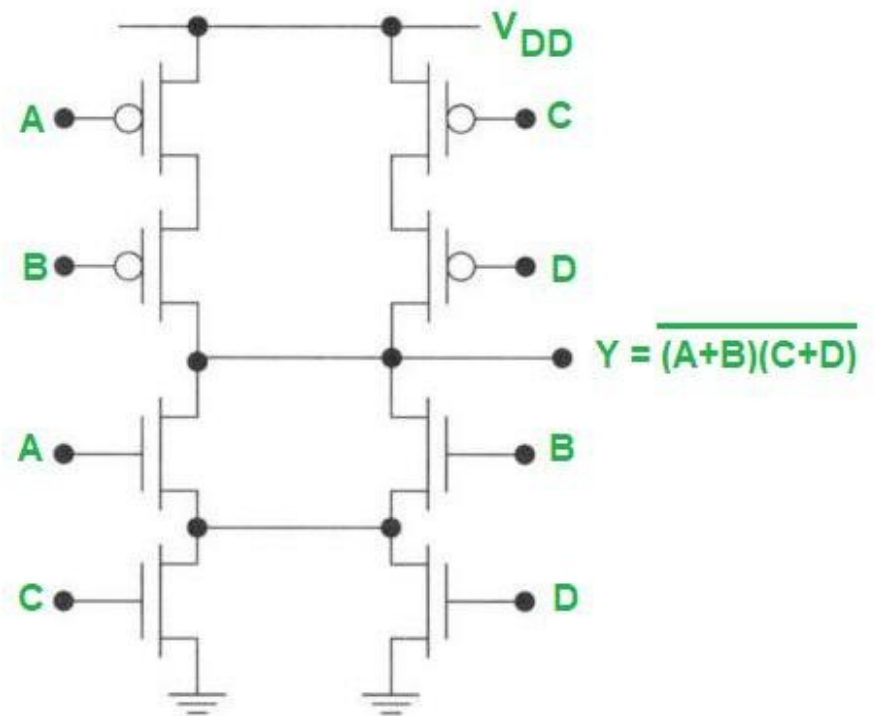
CMOS OR

CMOS logic



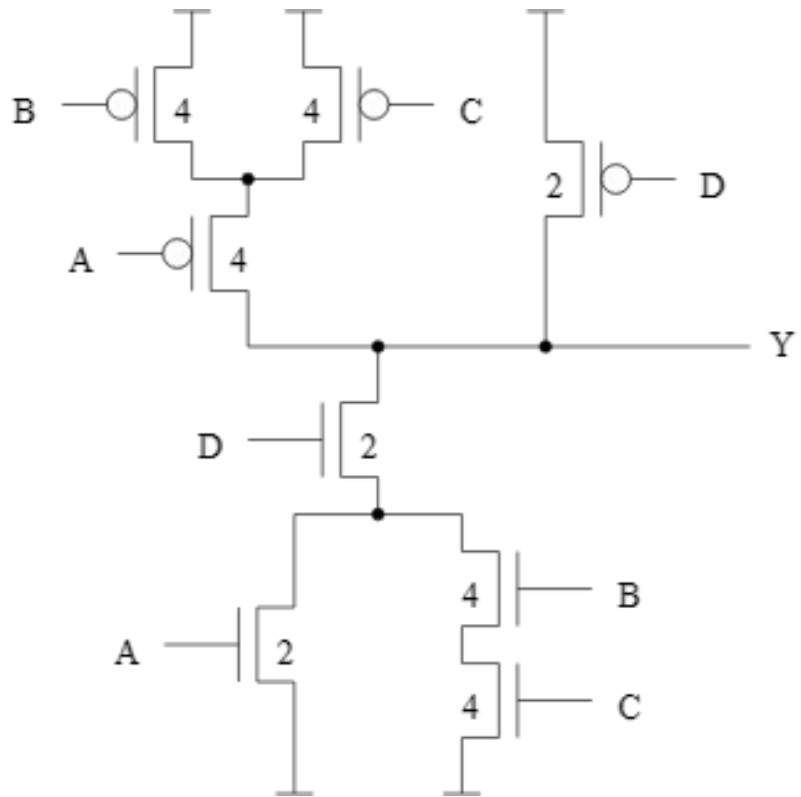


AOI Gate



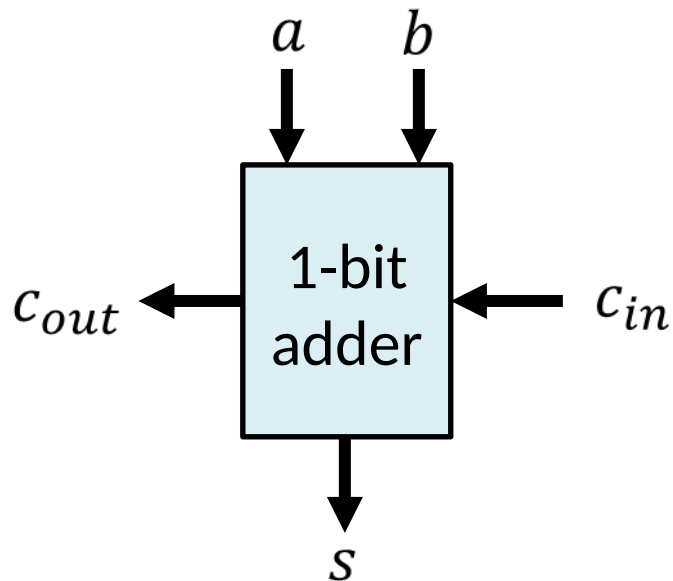
OAI Gate

$Y = ?$

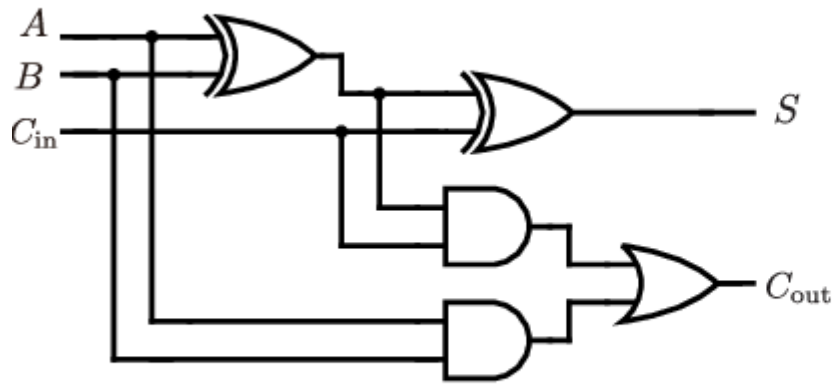


Adder

$$\begin{array}{r} \text{01010100} \\ + \text{00110101} \\ \hline \text{10001001} \end{array} \qquad \begin{array}{r} 84 \\ + 53 \\ \hline 137 \end{array}$$

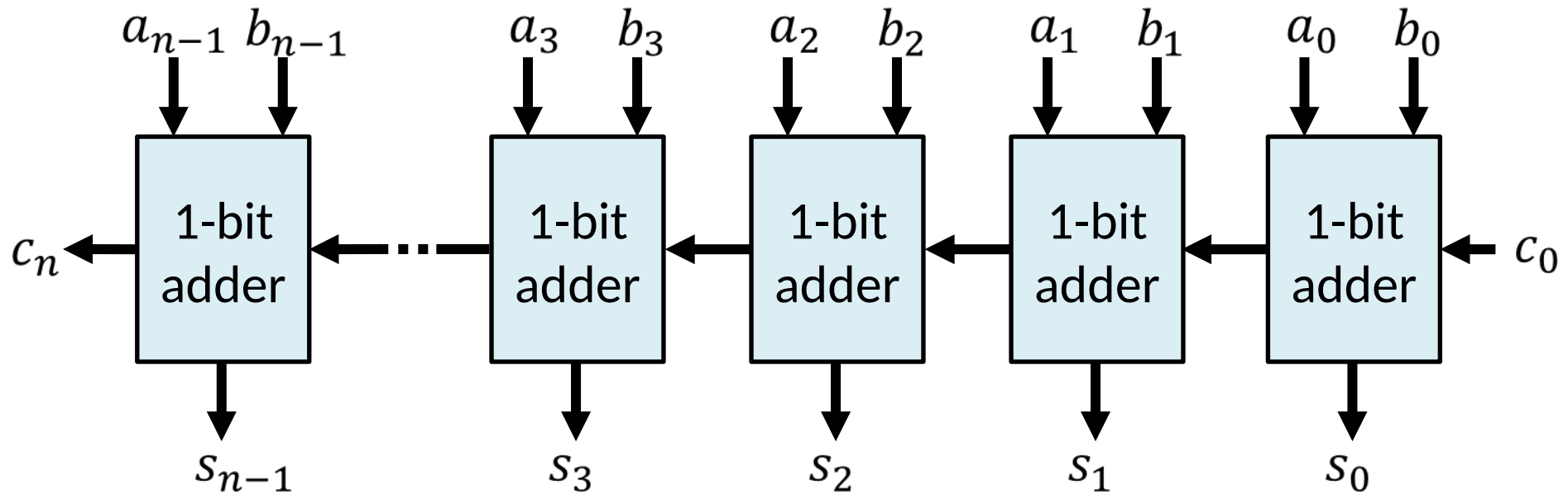


Adder

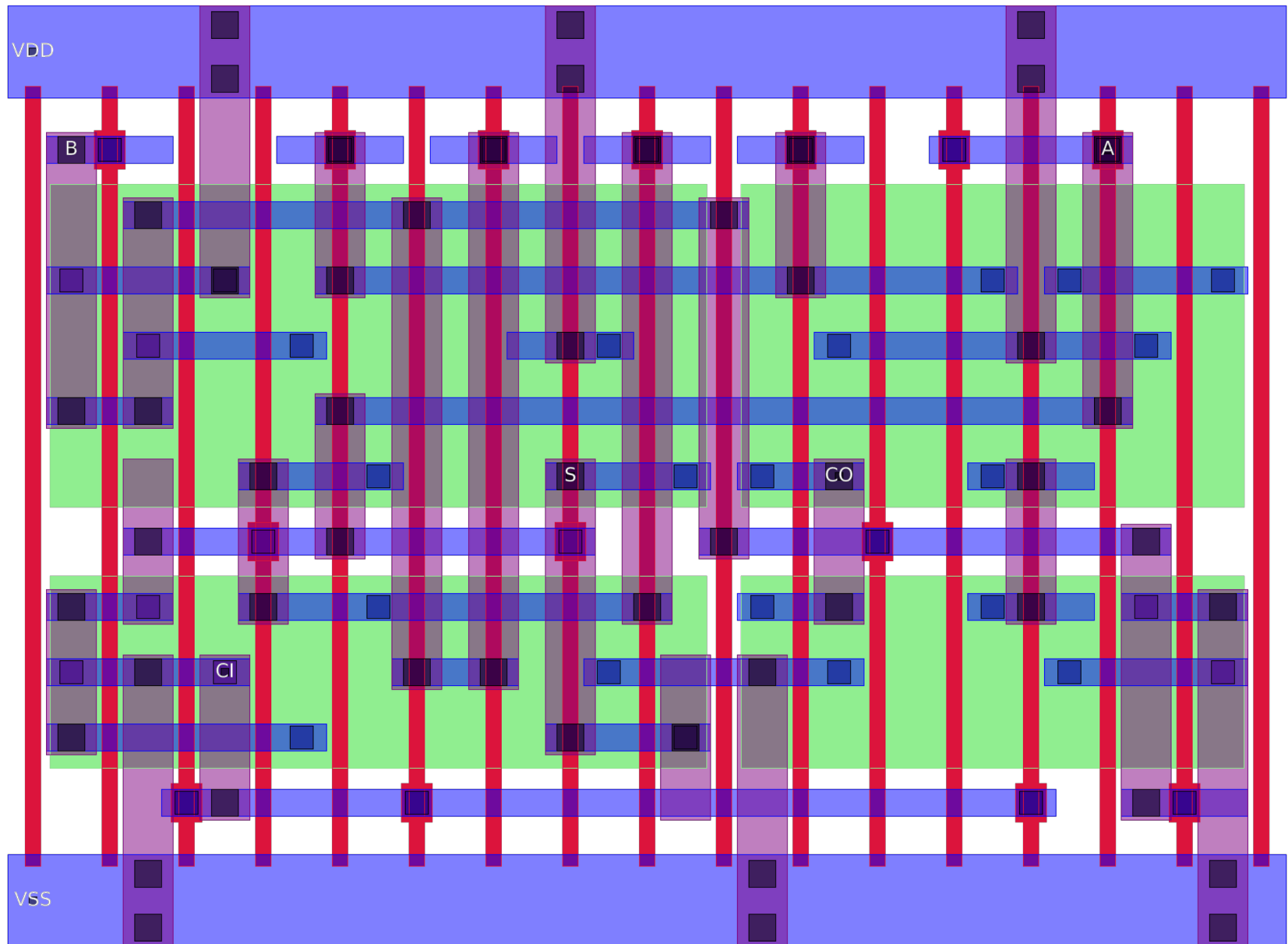


Inputs			Outputs	
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

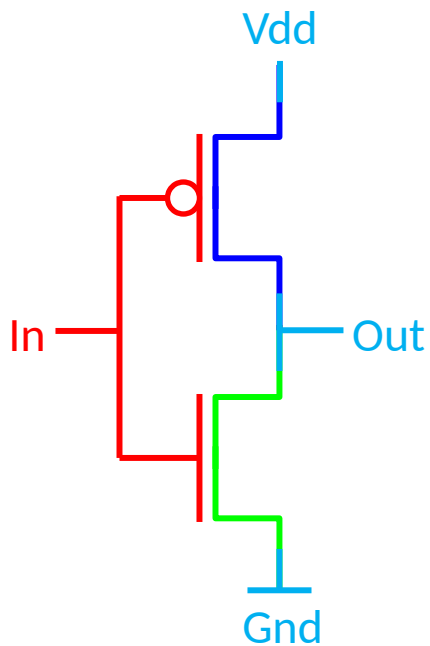
n -bit adder



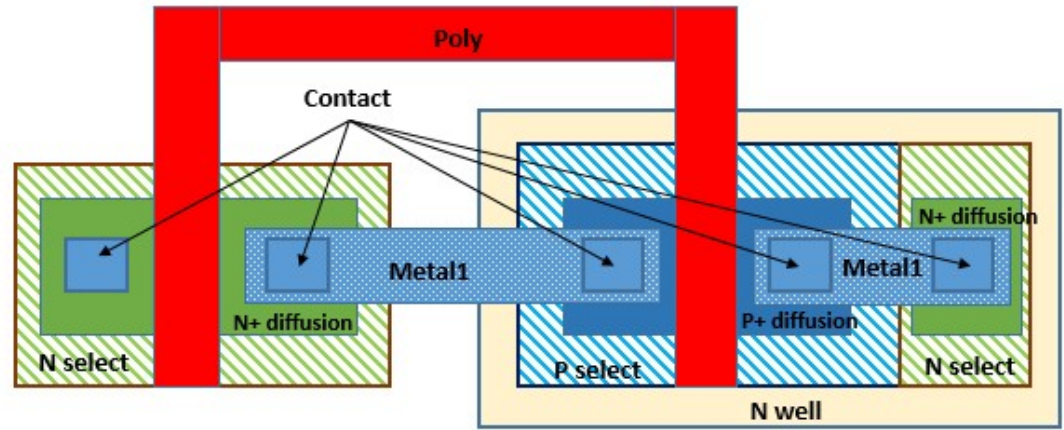
Layout of a 1-bit adder



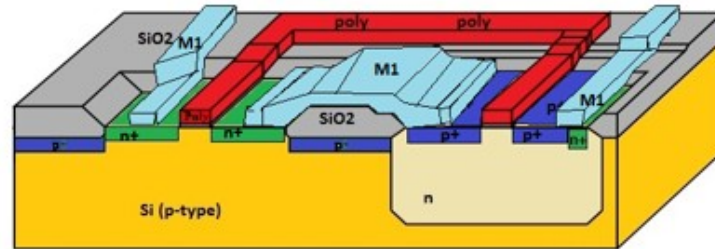
The inverter: layout



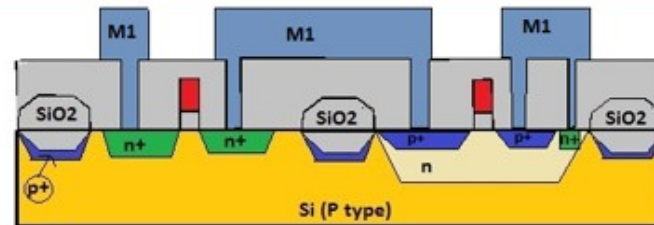
Top View

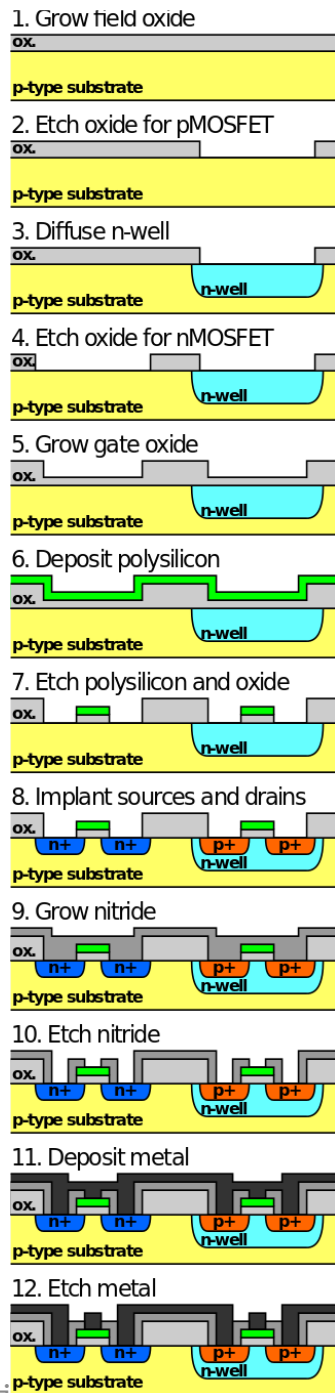


3D View



Side View

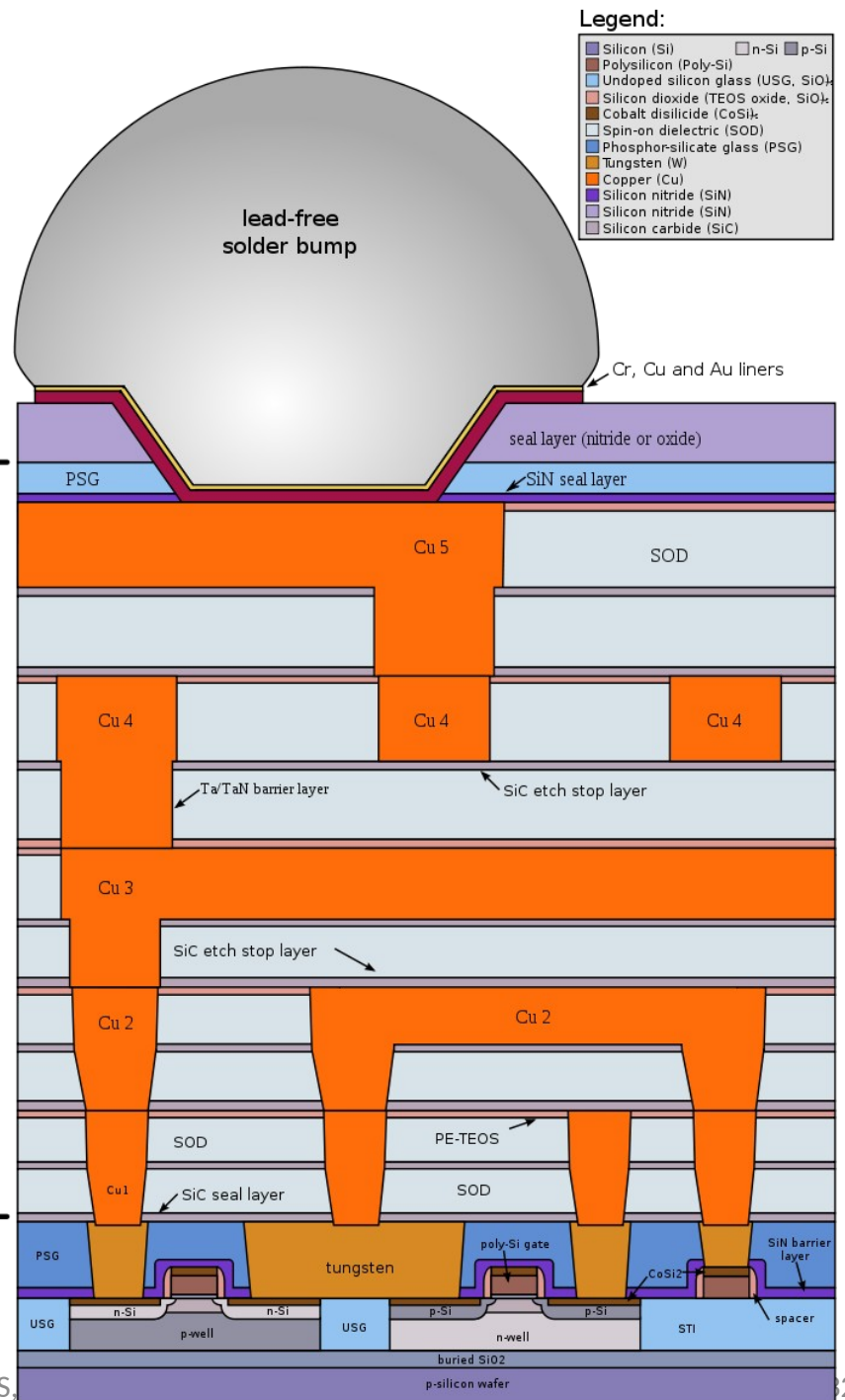


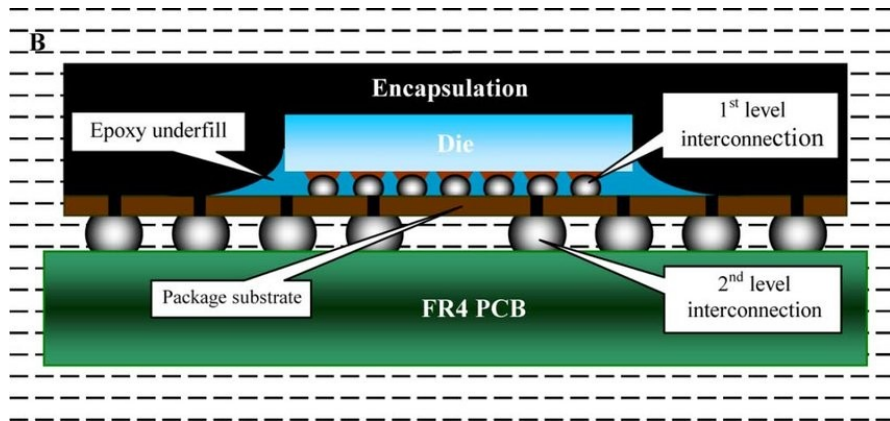


back-end /
"Advanced Packaging"

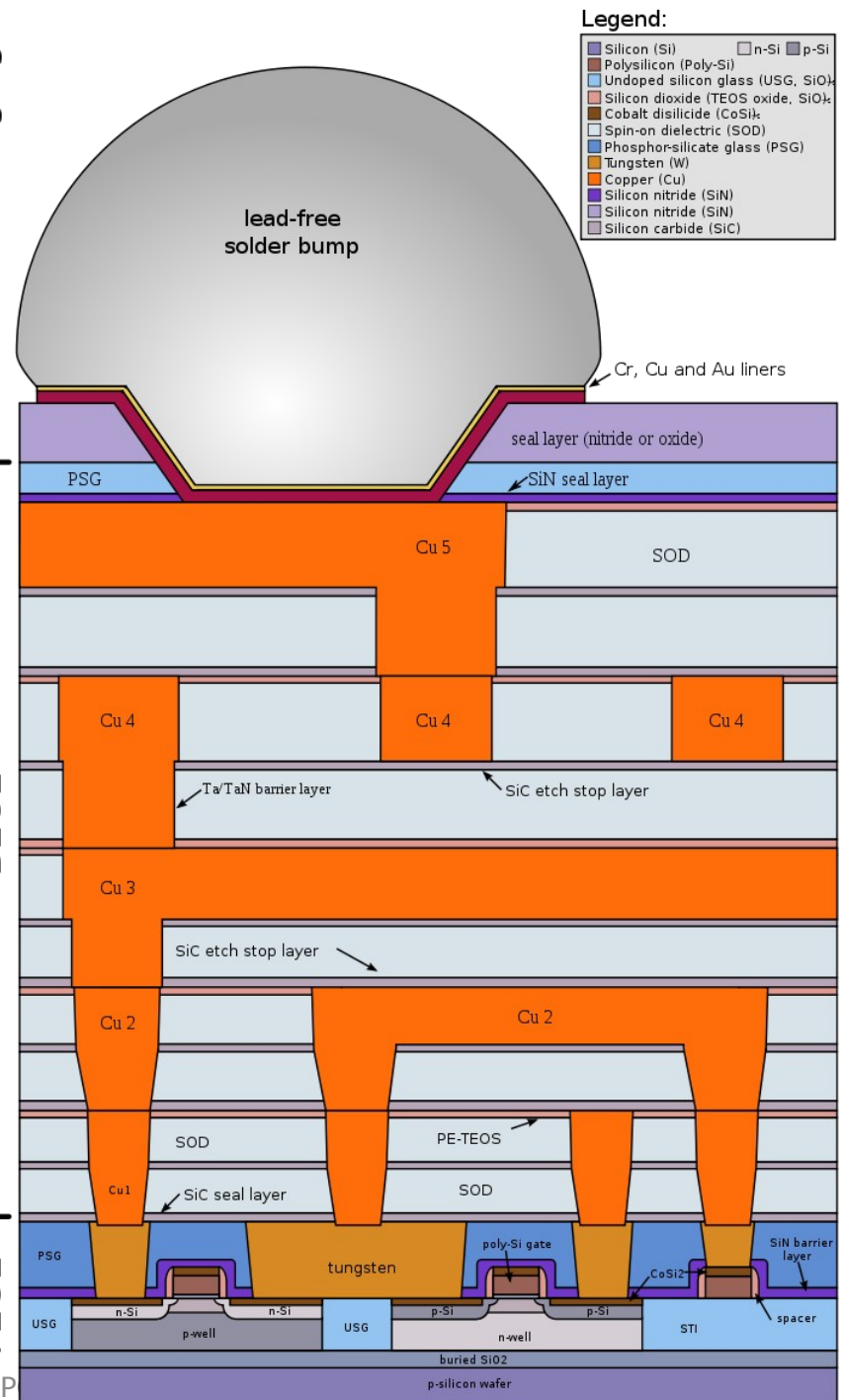
BEOL

front-end
FEOL





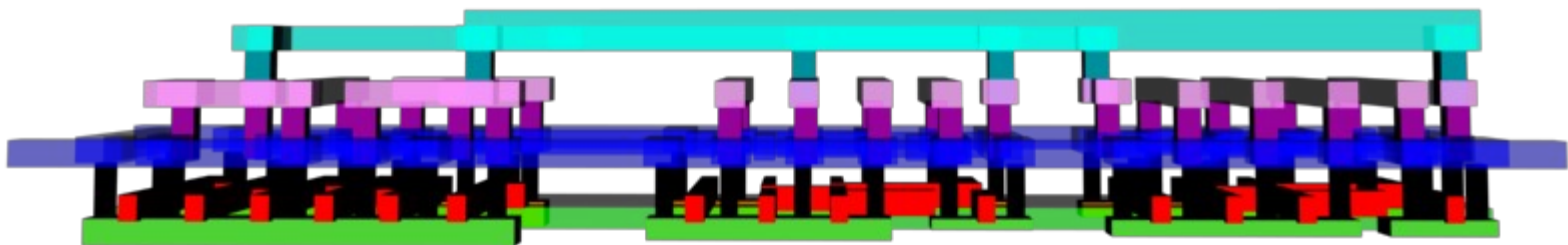
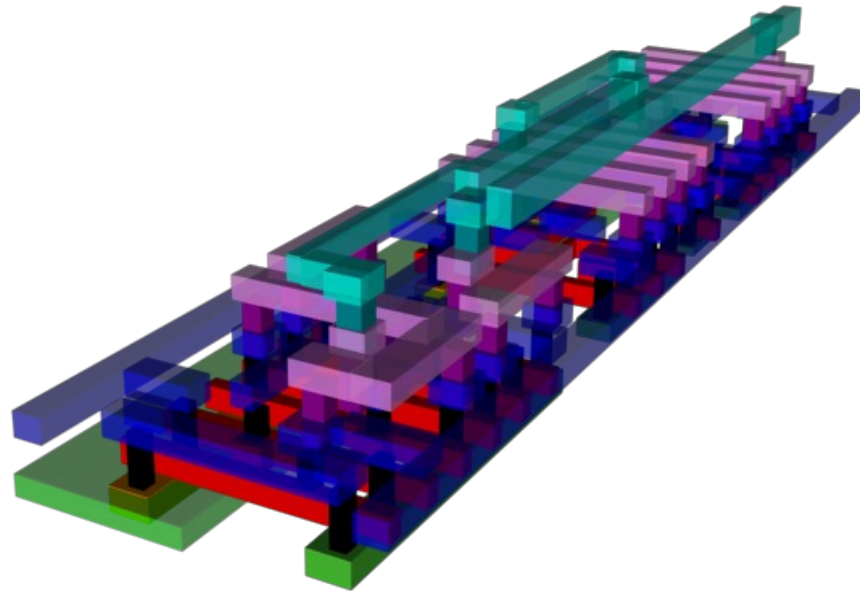
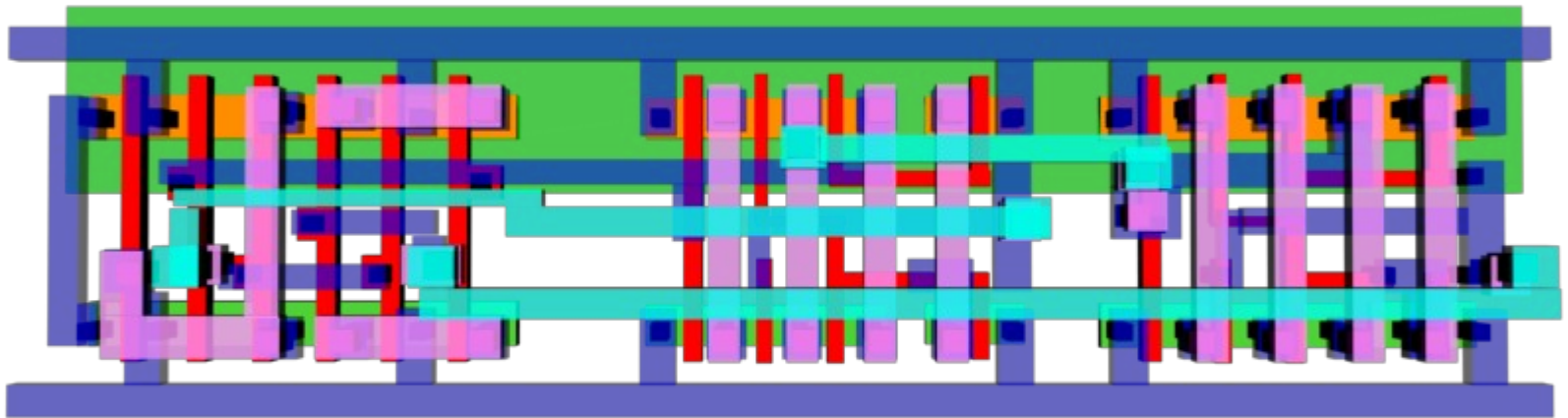
back-end /
"Advanced Packaging"

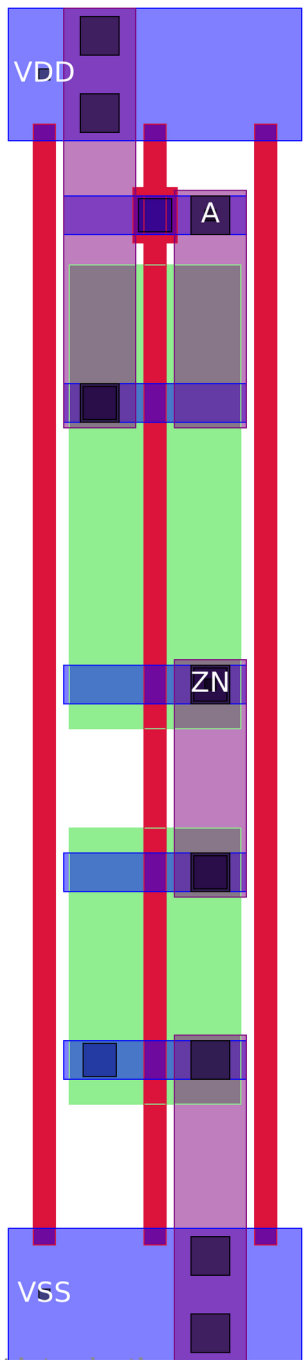


front-end

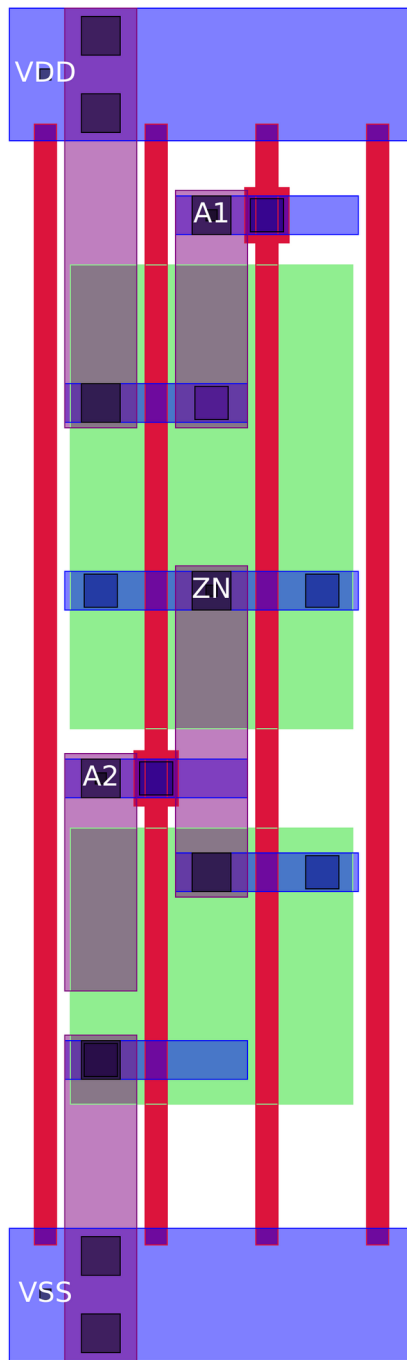
BEOL

FEOL

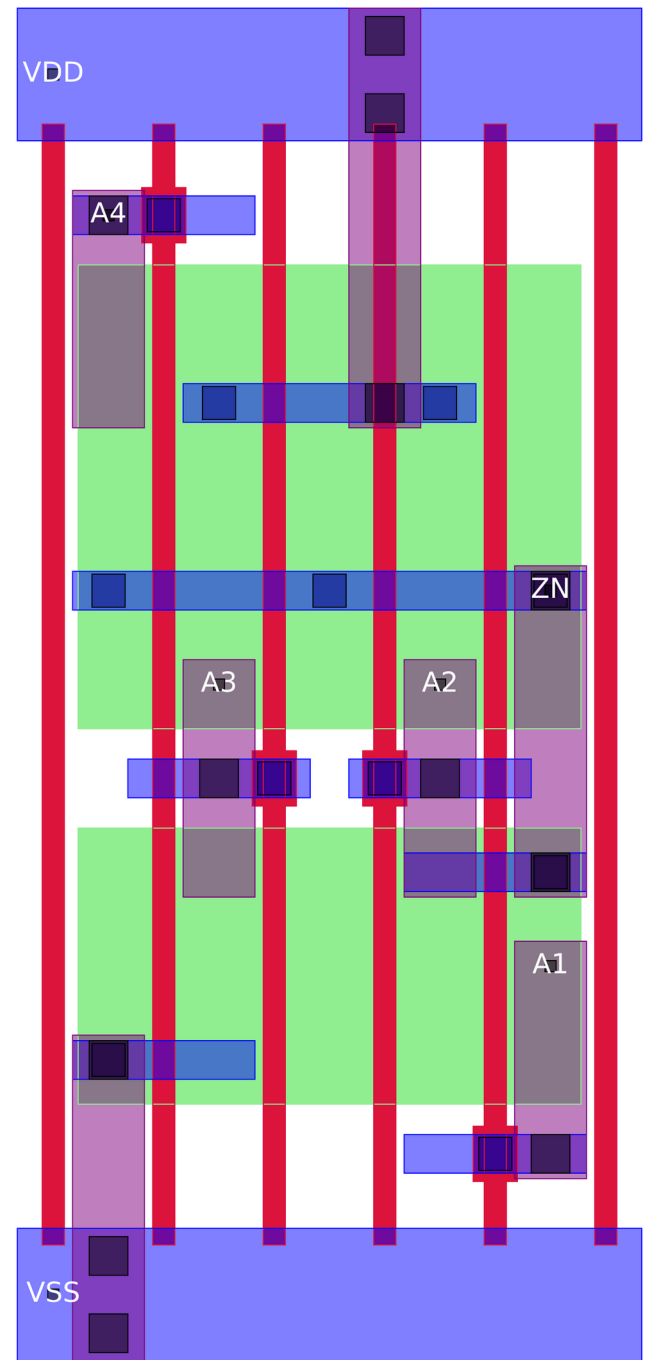




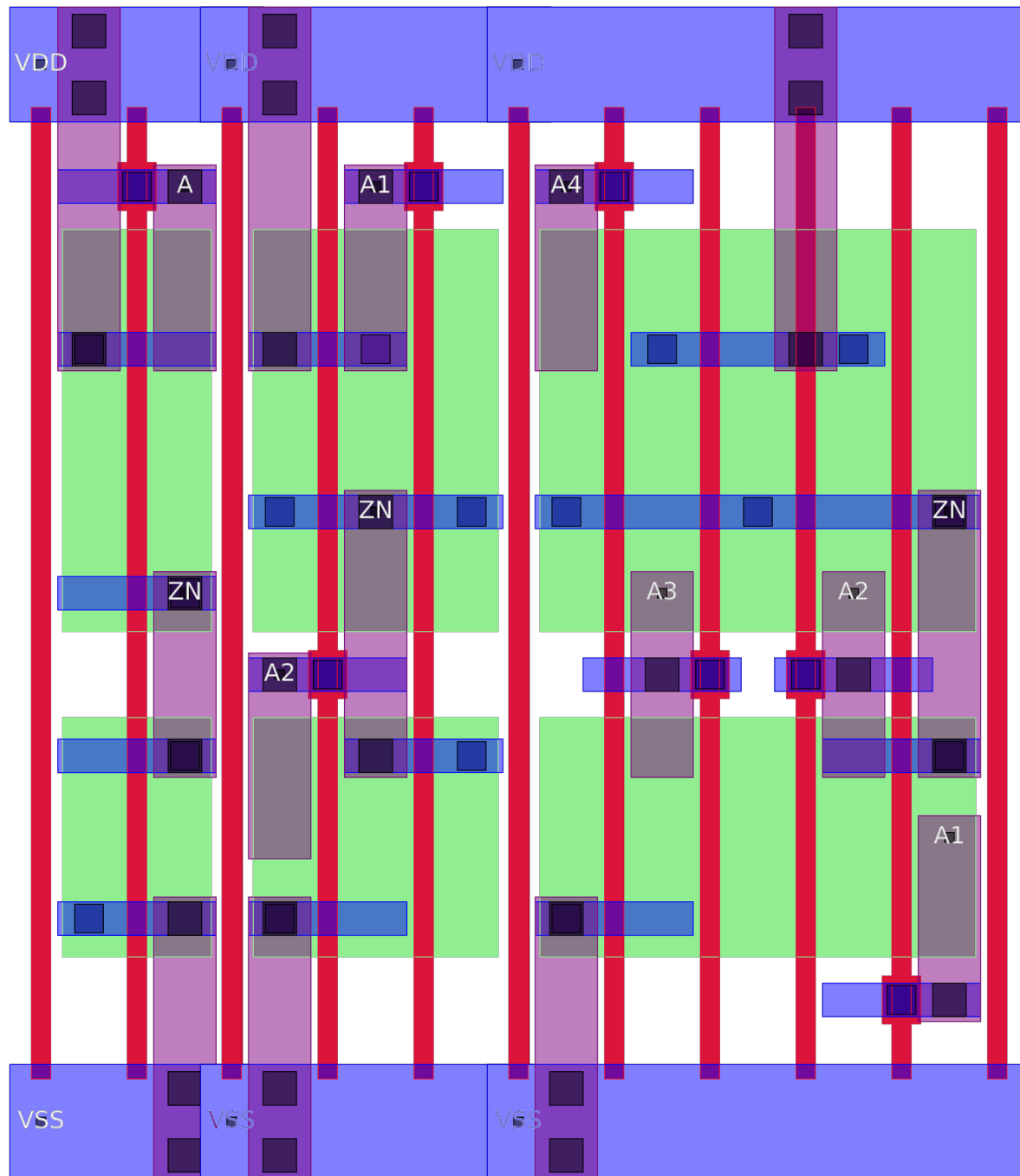
AVLSI introduction

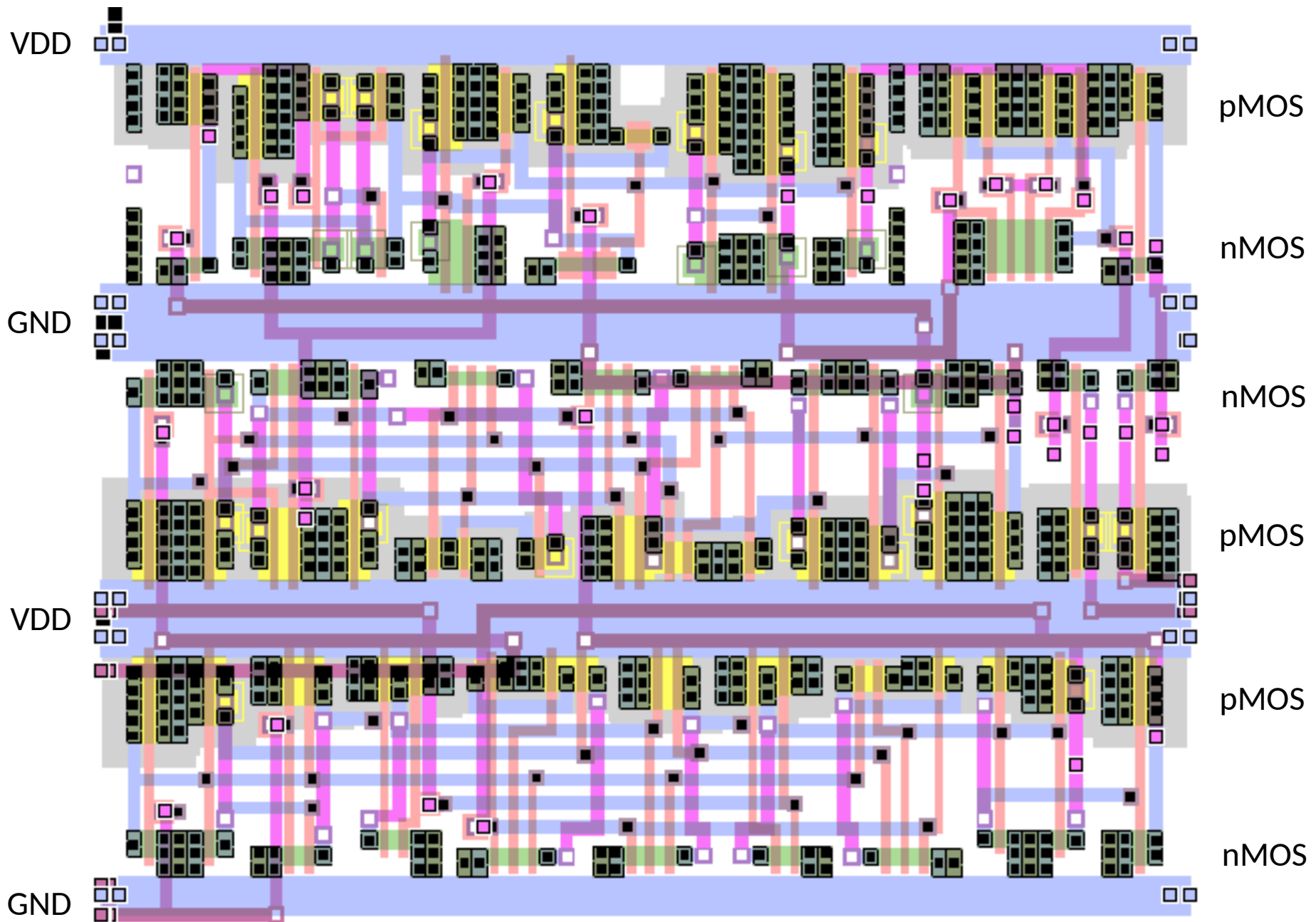


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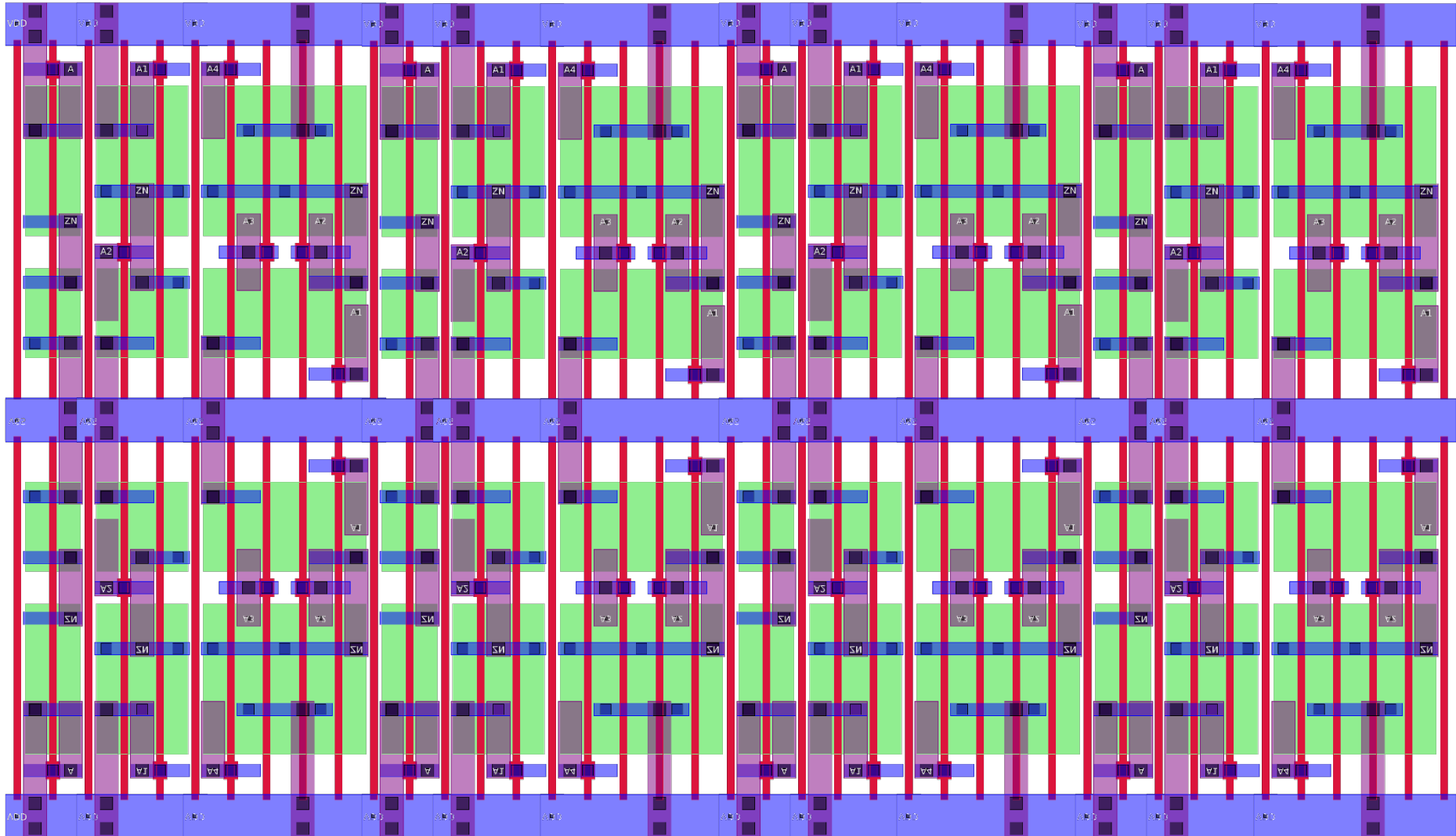


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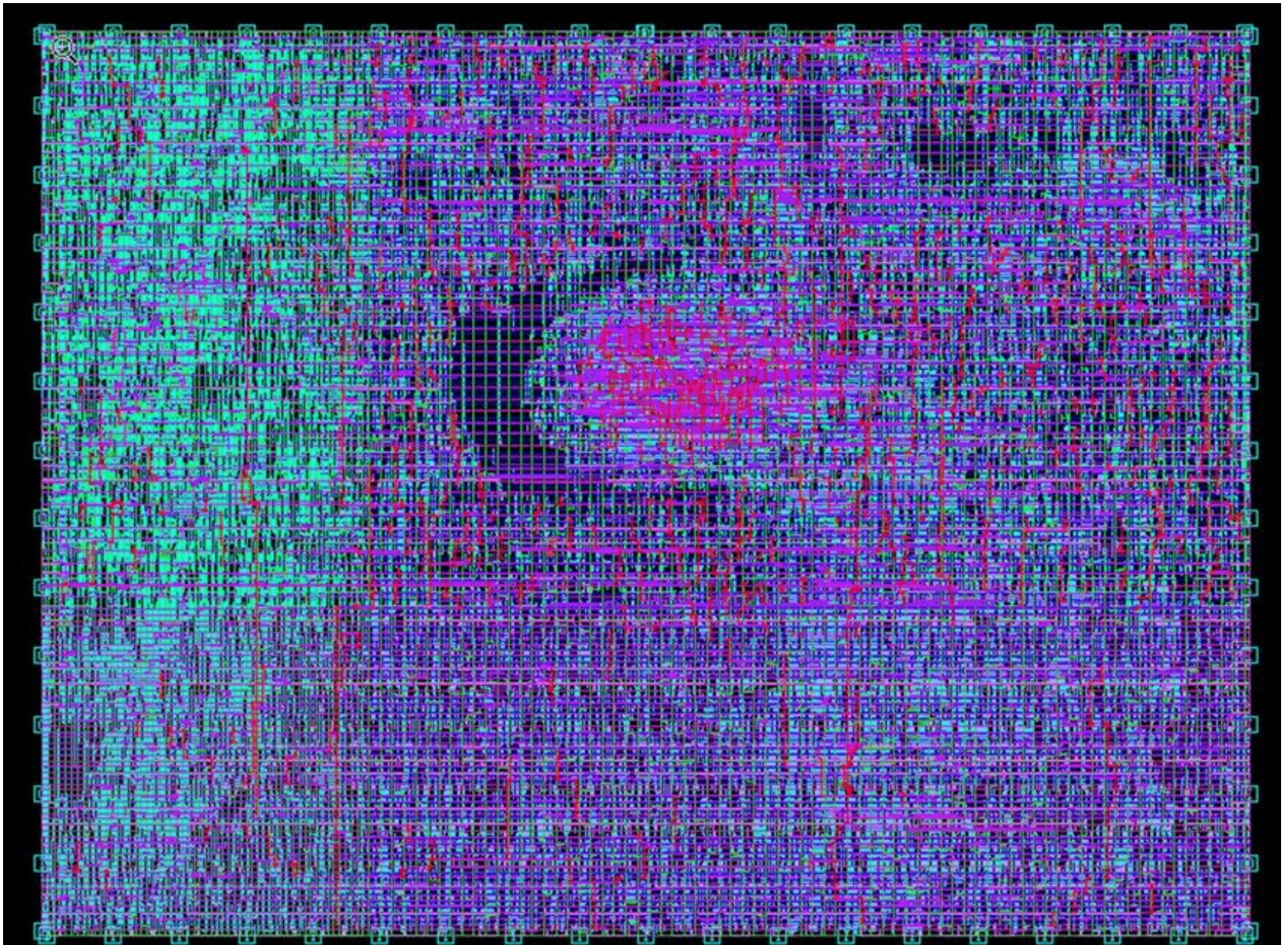




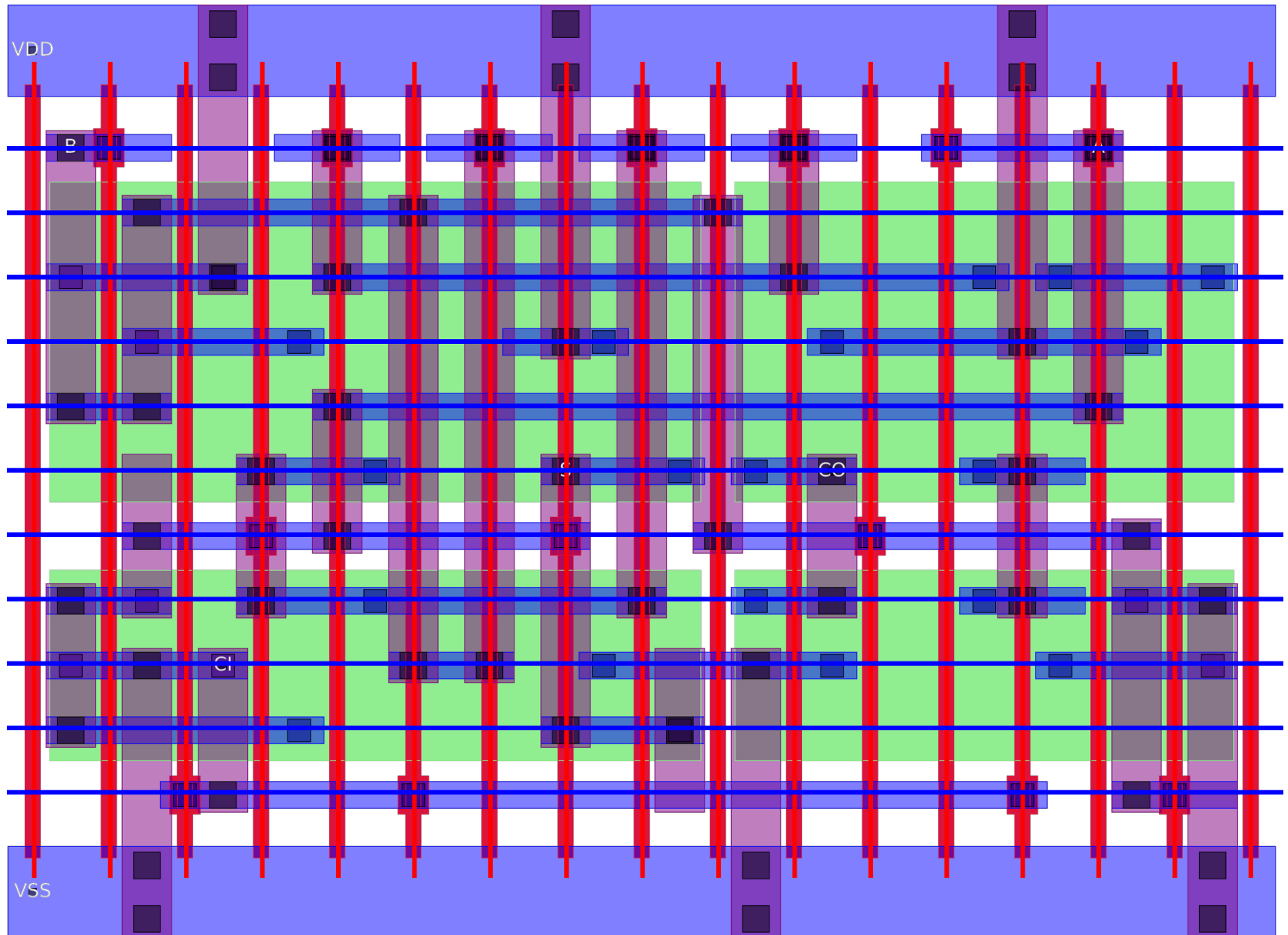
Gridded layout



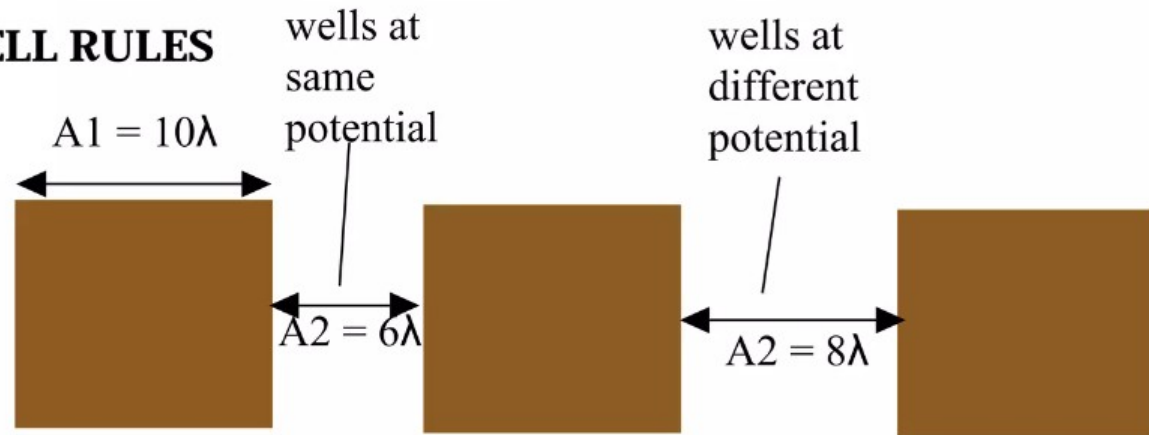
After routing



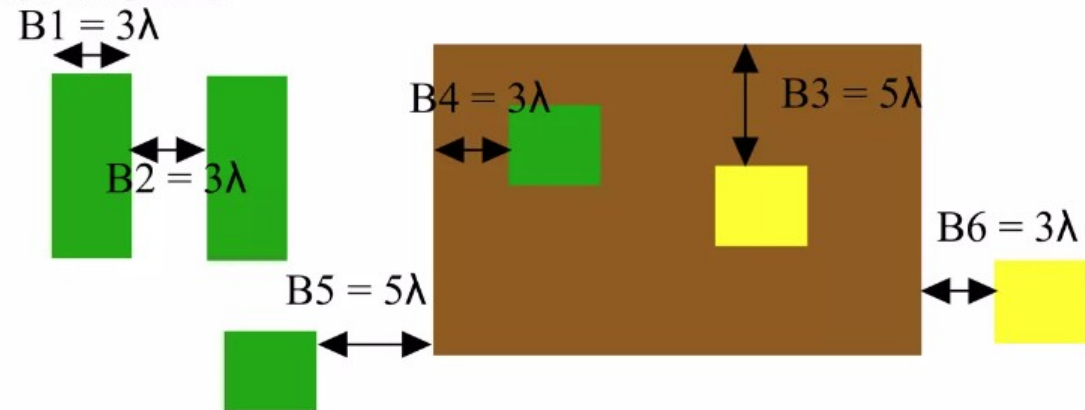
Gridded layout



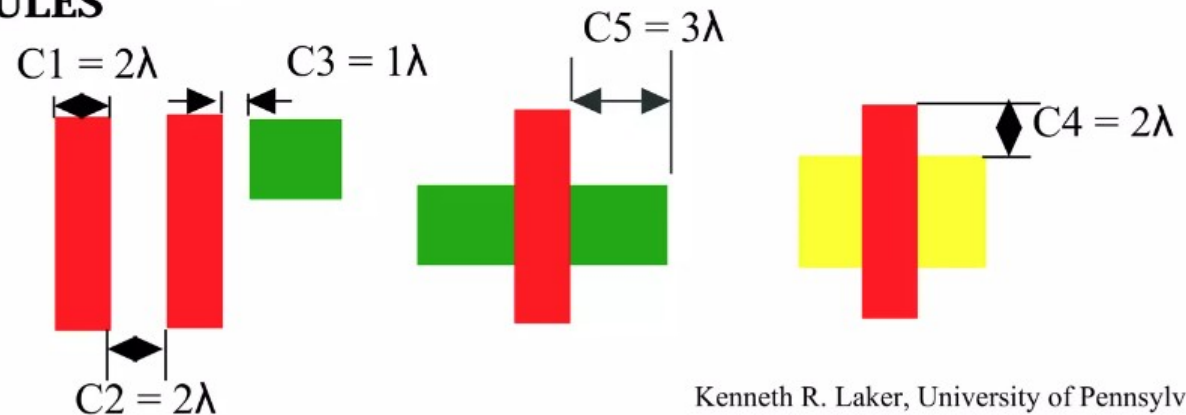
A- N-WELL RULES



B- ACTIVE RULES



C- POLY RULES



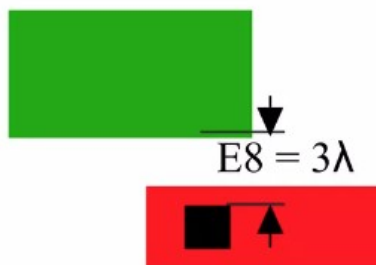
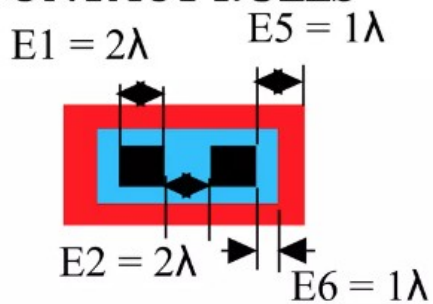
COLOR LEGEND

	n-Well
	p-Well
	n ⁺
	Polysilicon
	p ⁺
	Gate Oxide
	Field Oxide
	Metal 1
	Metal 2
	Metal 3
	Contact/via

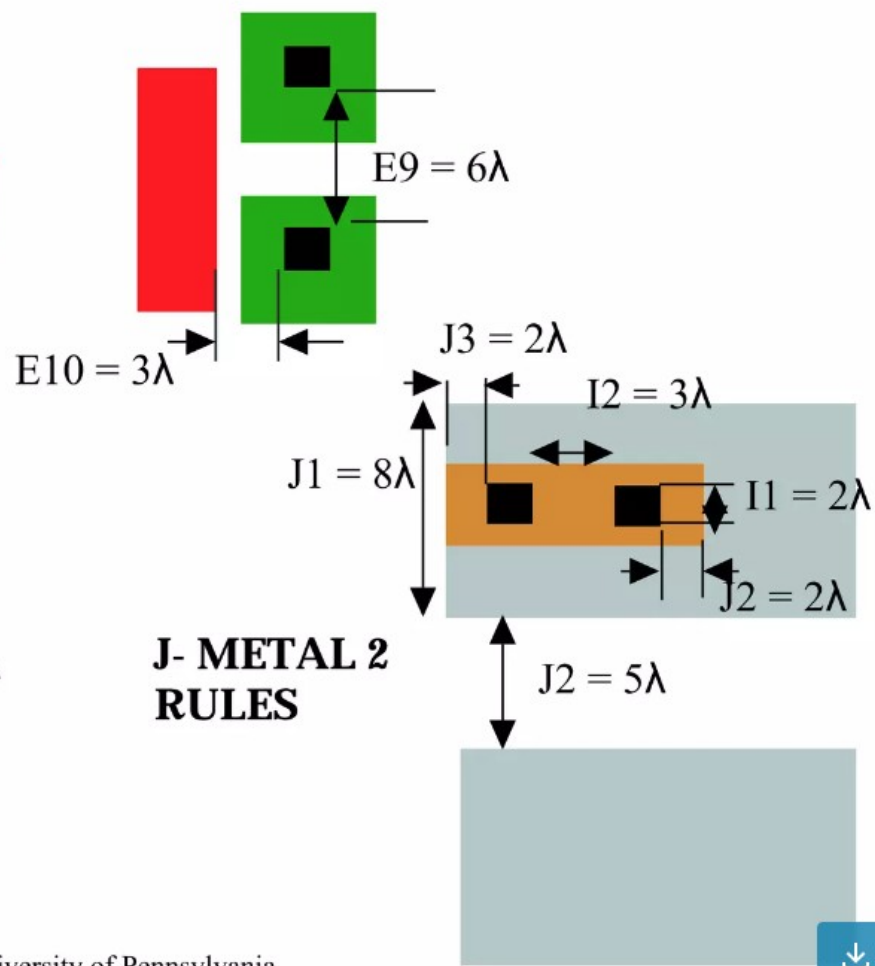
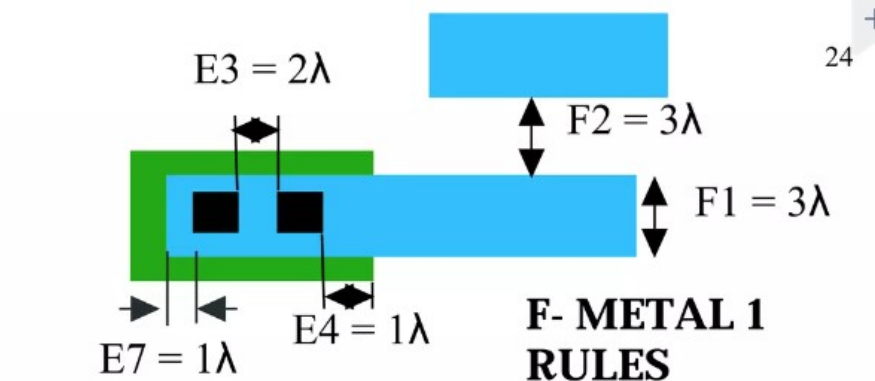
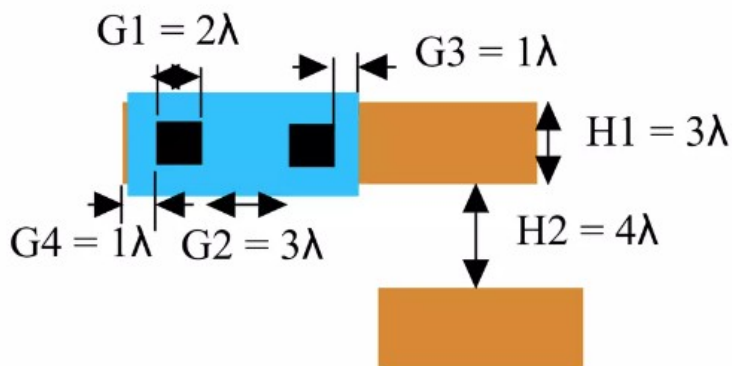
COLOR LEGEND

	n-Well
	p-Well
	n ⁺
	Polysilicon
	p ⁺
	Gate Oxide
	Field Oxide
	Metal 1
	Metal 2
	Metal 3
	Contact/via

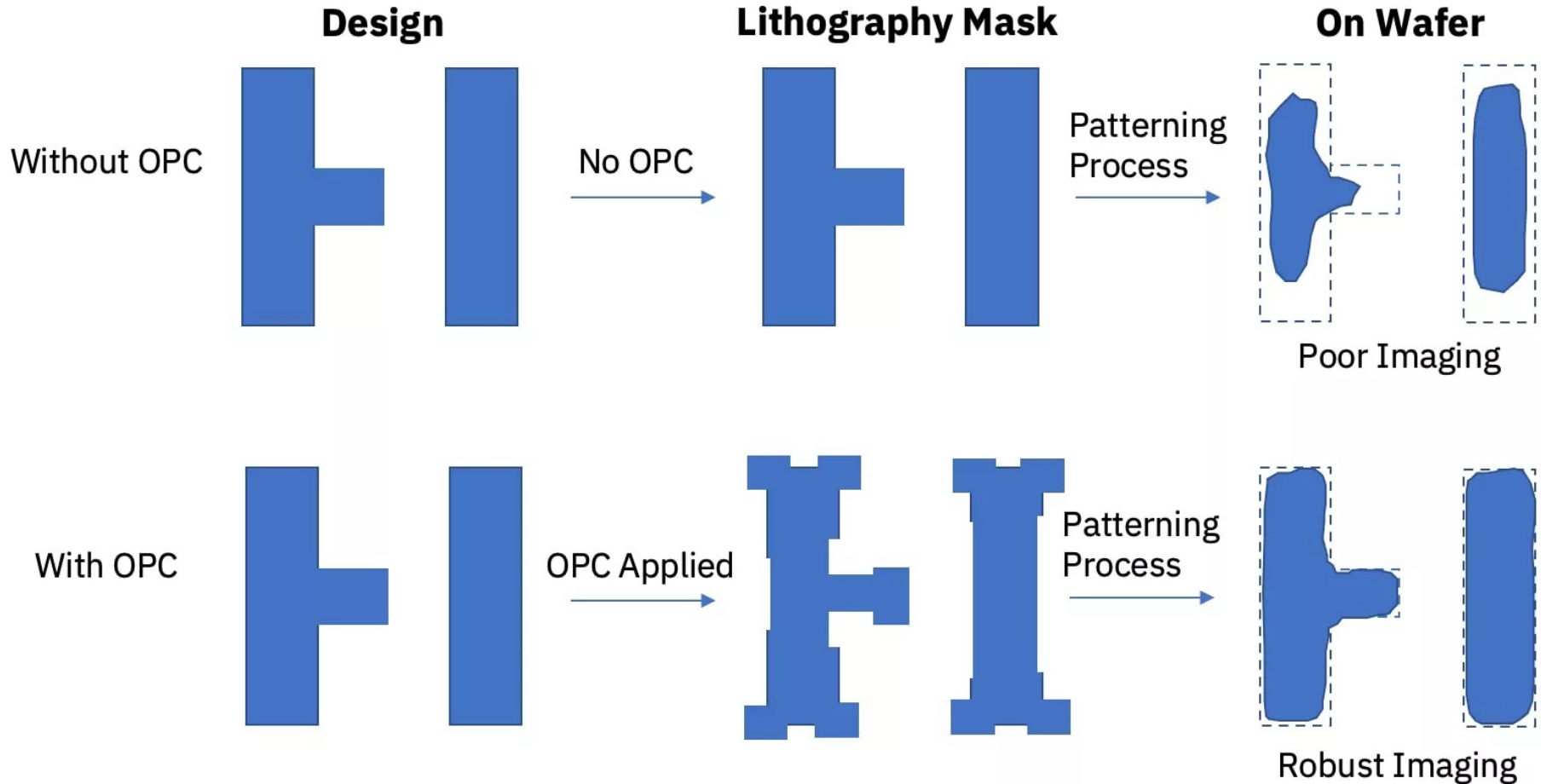
E- CONTACT RULES



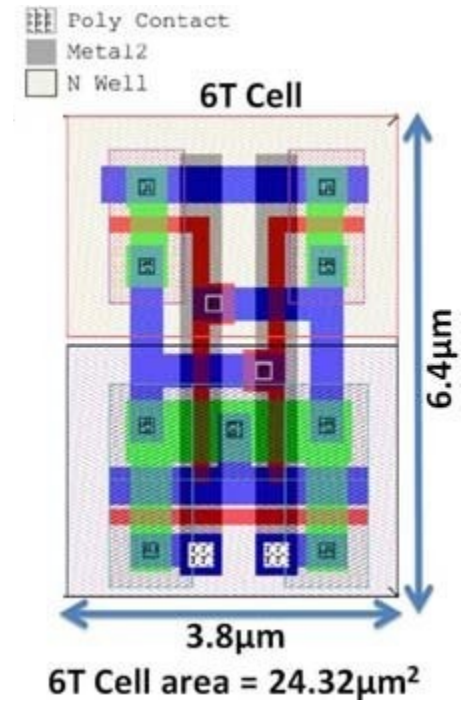
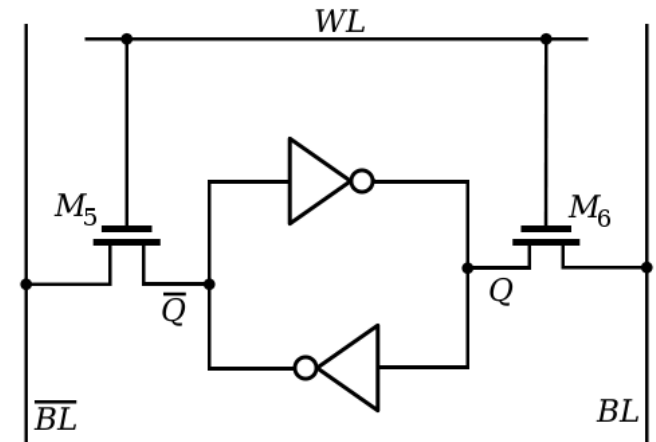
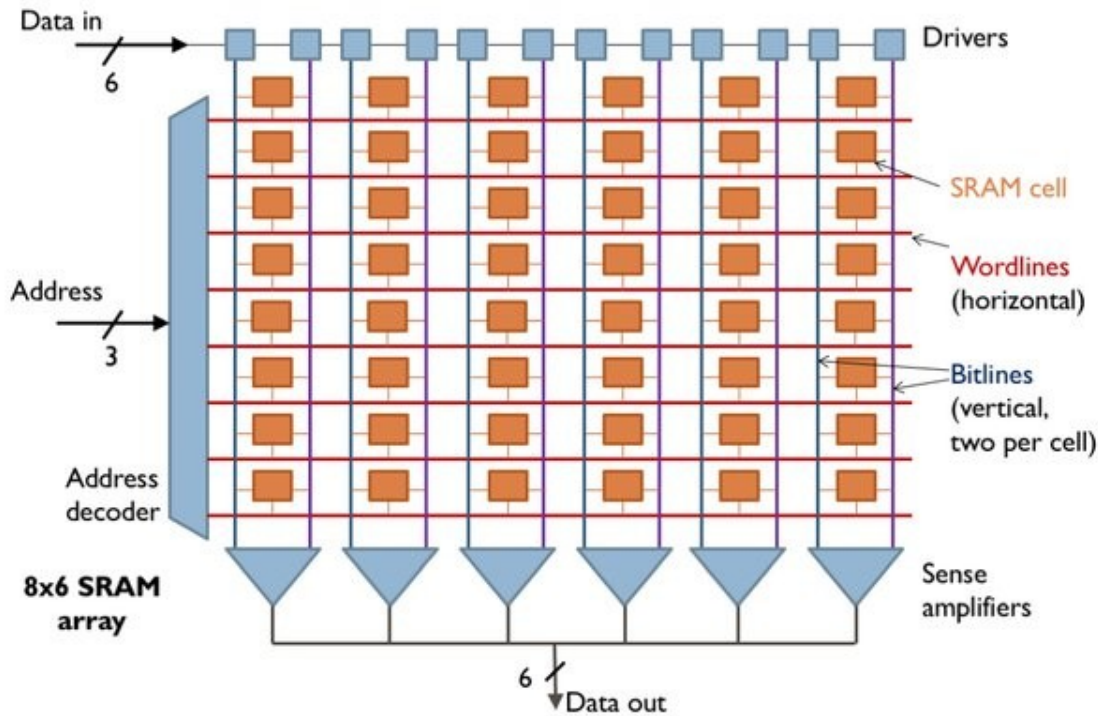
G- VIA RULES



Optical Proximity Correction (OPC)



Static RAM (SRAM)



Introduction: CMOS videos

- [How the transistor works](#)*
- [CMOS transistor](#)
- [Photolithography](#)*
- [How Billions Of Microchips Are Made Inside The World's Most Advanced Factory](#)*
- [How Are Microchips Made?](#)
- [VLSI fabrication process \(Global Foundries\)](#)
- [Semiconductor Manufacturing Process \(Samsung\)](#)
- [Fabrication of Integrated Circuits \(Intel\)](#)
- [Intel: The Making of a Chip with 22nm/3D Transistors](#)